17 Mediators and Mechanisms of Inflammatory Pain

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After tissue injury or exposure of the organ to a foreign body or infectious entity, a complex cascade of cellular and humoral events is initiated. These events, observed in virtually every organ system, including skin, muscle, meninges, dentition, bone, and visceral tissues, fall broadly under the rubric of "inflammation." In general, this cascade serves to protect and maintain the functional integrity of the systems in the face of insult. The process activates the immune system and directs chemotactic agents, neutrophils, and mononuclear cells to migrate from the vascular bed to the injury site. The cardinal clinical signs of this process, *rubor/calor* (increased local blood flow) and *tumor* (swelling secondary to local plasma extravasation) are manifested to varying degrees in all of these tissues. The fourth cardinal sign, *dolor* (pain), typically accompanies such cascades. The biological processes whereby these inflammatory cascades serve to initiate and sustain a pain state are the focus of this commentary.

INFLAMMATORY PAIN PHENOTYPE

The acute application of a thermal or mechanical stimulus of such intensity as to *potentially* produce tissue injury will typically evoke a somatic escape response (e.g., a withdrawal of the stimulated limb) and an autonomic response (e.g., hypertension and tachycardia), a syndrome classically referred to by Sherrington as a nociceptive reflex [1]. These acute responses typically display four characteristics: (i) the magnitude of these responses varies directly with stimulus intensity; (ii) the latency varies inversely with stimulus intensity; (iii) the focus of the response is referred to as the specific site of stimulation (e.g., it is homotopic); and (iv) in the absence of tissue injury, removal of the acute stimulus results in a rapid attenuation of the sensation and the attendant behaviors.

In the face of local tissue injury and inflammation, a distinct pattern of aversive sensations is reported in the human and parallel effects are noted in nonverbal organisms as defined by the attendant behaviors. This behavioral phenotype typically characterized as a dull, throbbing, aching sensation in the injured area of skin, soft tissue, or joint. Inflammation of the linings of the abdominal or thoracic cavity (e.g., pleura or pericardial sac) or visceral organs such as kidney, pancreas, or gastrointestinal (GI) tract will lead to ongoing pain sensations that are referred to specific somatic dermatomes, for example, where coronary ischemia is referred to the left arm and shoulder or an appendicitis to the lower right abdomen in humans [2]. Such referred pain is commonly reported to be burning or cramping in character. As discussed further later, in nonverbal models, the ongoing pain sensation is intuited from guarding of the limb, reduced weight bearing, impaired ambulation and ongoing vocalization, both sonic and ultrasonic [3,4]. Abnormal grooming or licking of the affected body part is also noted. In the face of visceral inflammation, abnormal postures such as stretching of abdomen and/or increased abdominal tone may be present [5]. In the face of injury and inflammation of somatic or visceral tissues, there is typically evidence of enhanced autonomic activity (e.g., increased blood pressure and heart rate, pupillary dilation) [6].

Facilitated Pain States

Application of evocative stimuli to the injury site such as a mechanical distortion or thermal stimulation of the tissue or flexion of the joint in a manner that is not normally considered to be noxious will lead to reports of pain sensations. This lowered magnitude of stimulus intensity required to elicit an aversive response to a stimuli applied to the injury site is referred to as primary hyperalgesia [7]. In the face of local injury

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and inflammation, it is also appreciated that stimuli applied to injured tissue will be responded to in an exaggerated fashion. Thus, low-intensity mechanical or thermal stimuli applied to the skin, modest flexion of the inflamed joint, or moderate distention of the gastrointestinal (GI) tract will lead to an augmentation of the ongoing behavioral signs [8]. In addition, low-intensity stimulation applied to regions *adjacent* to the injury site may also produce a pain condition. This sensitization of adjacent noninjured tissue to low-intensity stimuli is then referred to as 2° hyperalgesia or allodynia [9]. In human experimental models, these response components have been readily identified with a focal injury applied to the forearm or hand [10]. In summary, this property of sensory "sensitization" with local injury is generalized to virtually all organ systems. Thus, inflammation of the skin by sunburn leads to extreme sensitivity to warm water, pleural inflammation may lead to pain secondary to chest expansion (inhalation), irritation of the cornea lead to pain secondary to eyelid closure. In the case of inflammation of the bowel, light touch applied to the abdomen reveals reported "tenderness" and an increased abdominal muscle tone [11,12].

Preclinical behavioral models and inflammation/injury initiated pain states.

Defining the role of inflammatory mechanisms in "pain" or nociception depends on the effects of manipulating these various components of the pain behaviors otherwise generated by the inflammatory state. As reviewed earlier, such pain states may be composed of spontaneous behavior and changes in thresholds that occur secondary to the injury (e.g., hyperalgesia). Animal models involving local tissue injury or inflammation at specific target sites, for example, the paw or the GI tract, has been systematically used to assess pain behavior over time after injury. One such material carrageenan (linear sulfated polysaccharides extracted from seaweed) has been classically used as a model of acute inflammation [13]. As shown in Figure 17.1, the intraplantar injection of carrageenan into one hind paw will lead to (i) a time-dependent inflammatory response in the injected paw; (ii) a corresponding decrease in the magnitude of the mechanical (or thermal) stimulus required to evoke a withdrawal of the injured hind paw; and (iii) a reduced response latency. The latency/mechanical threshold of the contralateral paw in this model is little changed. Here the thermal escape latency is measured using a system that permits targeting the radiant thermal stimulus independently to either hind paw (a Hargreaves device) [14], while the mechanical threshold is determined using von Frey filaments [15]. This model demonstrates the

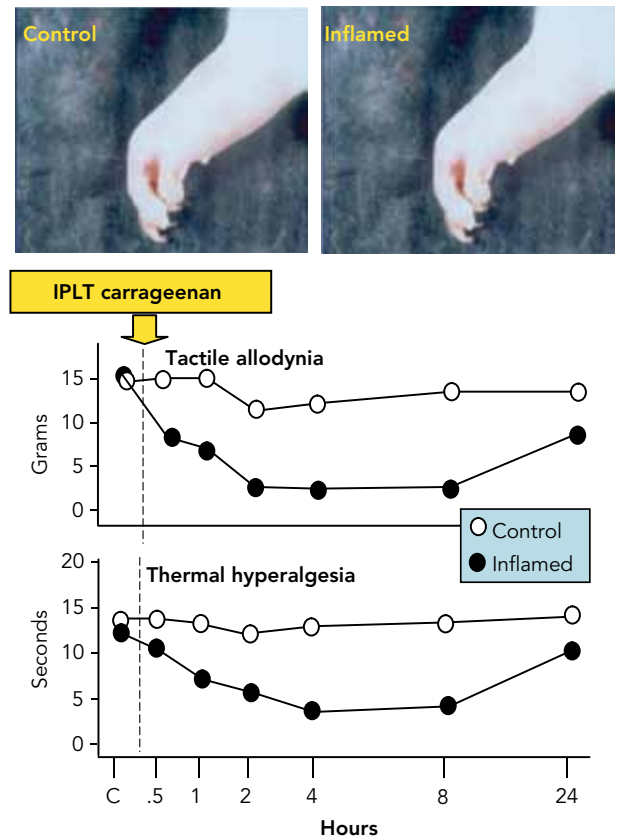


Figure 17.1. Unilateral intraplantar injection of carrageenan will initiate a potent inflammatory response (top). Measuring the thermal escape latency on the paw contralateral to the carrageenan shows ho change in the tactile thresholds (middle) or thermal escape latencies (bottom). In contrast, over the 24 hr interval, there is a robust fall in tactile and thermal thresholds.

prominent and long lasting thermal and mechanical hyperalgesia of this paw. Similar behavioral effects can be observed after the delivery of a variety of proinflammatory treatments such as (i) complete Freund's adjuvant (evoking a disseminated autoimmune response to a joint or connective tissue antigen) [16,17]; (ii) local injuries, for example, focal thermal injury [18]/incision [19] of the paw; (iii) intraarticular injection of irritants (carrageenan-kaolin [20]); (iv) delivery of products that deliver or initiate antibodies against joint components (such as collagen [21]); (v) inflammation of the GI tract produced by the instillation of products such as trinitrobenzene sulfonic acid (TNBS) (a hapten that induces colitis [22]) or iodoacetamide [23]; (vi) bladder inflammation (cystitis) as produced by instillation of irritants such as cyclophosphamide [24]; or (vii) pancreatitis produced by dibutyltin dichloride (DBTC) [25]. All of these treatments in various representative organ systems will commonly produce evident behavioral signs of ongoing discomfort and hypersensitivity. Inflammation or injury of visceral organs will typically

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TABLE 17.1. Classification of primary afferents

Fiber class	Size (μ)	Myelin	Velocity (m/sec)	DRG neuron	Transmitter	Characteristic epitopes	Typical stimuli	Psychophysical correlate of acute stimulus
Aß	12–20	Yes	>40–50	Lg	Glutamate	Neurofilament 200	Low threshold Mechanical/tactile	Light touch proprioception
Að	1–4	Yes	10 < 40	Lg	Glutamate	IB-4	Subpopulations: low threshold (thermal or mechanical) or high threshold (thermal or mechanical)	Warmth Light touch Sharp-stinging Thermal crush
C	0.5–1.5	No	<2	Small	Glutamate peptides (sP/CGRP)	TRPV1-r	Subpopulations: polymodal	Dull, throbbing burning

be accompanied by signs such as abdominal stretching and increases in the tone of the abdominal musculature (as measured by increased abdominal EMGs), tachycardia, and hypertension. In addition, GI inflammation is often accompanied by marked increases in the aversiveness displayed by the animals to the application of mechanical stimuli to the body surface (corresponding to referred pain sites and to distention of the GI tract with inflatable balloons [26]). In short, preclinical inflammatory models will yield in animals a constellation of behavioral and physiological effects that parallel the verbal pain descriptors of ongoing discomfort and evocative hyperpathia observed in conditions of acute and chronic inflammation in the human patient (see [12,27]).

PERIPHERAL COMPONENTS CONTRIBUTING TO INFLAMMATORY PAIN STATES

Primary Afferent

With few exceptions, the sensory correlates of the body are communicated to the brain via the activation of specific populations of primary afferents projecting to the medullary (head and neck), spinal (body regions below the neck), and dorsal horn (see [28,29]). In visceral pain, information may also be trafficked through cranial nerves such as the vagus (Xth nerve) [30,31]. The sensory afferents, the cell bodies of which are found in the trigeminal (head and neck) and dorsal root ganglion (below the neck), are subdivided into classes reflective of their size, state of myelination, and the modality of stimulation that most effectively results in activity in the associated afferent axon (Table 17.1). On the basis of psychophysical responses evoked by stimuli that can activate these respective populations of axons, specific categories of sensations have been affiliated with the information carried by these afferents. Conversely, in the absence of injury, the range of stimulus intensities that most effectively activate a population of afferents is typically reflective of the presumed role played by these axons in initiating a sensory state. Thus, light touch is typically associated with the large mechanically sensitive afferents (Aß) that display sensitivity to low threshold mechanical stimuli. Að axons form subpopulations which may respond to low- or high-intensity mechanical or thermal stimuli, while C fibers respond predominately to high-intensity stimuli of thermal, mechanical and, importantly, chemical modalities. On the basis of differential sensitivity to blockade and the marked differences in conduction velocity, it is considered that the high-threshold Aðs may carry the initial stinging sensation generated by an acute high-intensity stimulus (e.g., first pain), while the slower C-fiber mediates the delayed dull, throbbing, aching sensation (2nd pain) [32].

Initiation of Small Afferent Activity

The primary afferent terminals of peptidergic C fibers are morphologically characterized as "free nerve endings." These peripheral terminals are highly ramified in most tissues and frequently appear as a triad of terminal, capillary, and mast cells [33]. Under normal conditions, in the absence of injury or inflammation, sensory afferents, particularly those with high-stimulus thresholds, show little or no spontaneous activity. Mechanical and/or thermal stimuli applied to the peripheral terminal of the afferent is converted by transducer proteins on the terminal into an intensity-dependent depolarization leading to an

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TABLE 17.2. Afferent transducer proteins

Channel	Activating parameters	Chemical sensitivity
TRPV1	>43°C	Lipids, capsaicin H^+
TRPV2	>52°C	
TRPV3	>34–38°C	
TRPV4	>27–35°C	
TRPM8	<25–28°C	Menthol
TRPA1	<17°C	Mustard oil
ASIC	H+	Acid
P2X	ATP (mechanical?)	

Abbreviations: TRP, transient receptor potential; ATP, adenosine triphosphate.

intensity-linked increase in the discharge frequency of these afferents. The mechanical and thermal stimuli sensitivity of a given afferent fiber is defined by the constituent transducer proteins that are present on the afferent terminal (see Table 17.2). These transducer proteins are typically effectively activated over a narrow range of intensities of the respective stimulus modality. Increasing stimulus intensities typically lead to the activation of higher threshold transducers and an increasing cation permeability, all leading to a progressive depolarization of the terminal [34]. The local depolarization then leads to the opening of local voltage-sensitive sodium channels and this leads in turn to an increasing frequency of afferent spike generation. An important property of these gated channels is that they may also be activated by specific chemical entities such as the TRPV1, which is activated by capsaicin and the TRP M8 channel by menthol. These activities parallel the psychophysical elements that such stimuli normally elicit, for example, sensation of heat and cold, respectively. Table 17.2 summarizes other such chemical correspondences.

Conduction of Sensory Afferent Activity

Voltage-sensitive sodium channels mediate the conducted potential. Cloning shows the presence of multiple populations of sodium channels. These multiple sodium channels have been identified based on structure (NaV 1.1-NaV 1.9), whether they are tetrodotoxin-sensitive or insensitive (TTX) and their activation kinetics. Some of the subtypes are geographically restricted in their distribution. Thus, NaV1.8 and 1.9 are largely limited to small primary afferents [35]. These NaV channels have several consensus sites that can be phosphorylated, serving to reduce their activation threshold and prolong their opening. As will be discussed later, these phosphorylation sites are important for regulating the excitability of the terminals by inflammatory products.

Primary Afferent Terminals and Inflammation

The preceding comments on the primary afferent and transduction mechanisms emphasized the transduction of acute high-intensity stimuli and an absence of spontaneous activity. As suggested earlier, increasing stimulus intensities (thermal, mechanical) will result in a monotonic increase in the discharge frequency of these axons. Local injury or inflammation in the environment of these peripheral terminals will, however, characteristically lead to (i) the appearance of high levels of ongoing (spontaneous) afferent activity; and (ii) a left shift in the stimulus discharge curve of these afferents such that there is a significant augmentation in discharge frequency at stimulus intensities that are of lower magnitude. Figure 17.2 summarizes such comments for thermal stimuli applied to the skin and for rotation of the joint. Here it is portrayed that the magnitude of spontaneous afferent traffic may mirror that which is otherwise observed after a moderate stimulus or an extreme joint rotation. It is appreciated that this exaggerated response reflects the inflammatory events that occur secondary to local tissue injury and inflammation.

The Local Inflammatory Milieu

Local tissue injury such as an incision or a burn results to varying degrees in a variety of distinguishable events that reflect activation of an innate immune response. These events, discussed in great detail in other chapters for various organ systems, may be heuristically subdivided as described in Figure 17.3. (i) Acutely, the stimulus serves to activate small high threshold sensory afferents. Such activation yields orthodromic and antidromic activity on the primary afferent. The antidromic activity (e.g., afferent traffic along collaterals toward the distal terminal away from the spinal cord) yields the local terminal release of sP and CGRP. These peptides result in a prominent local dilation of arterioles and plasma extravasation [36,37]. (ii) The damage to local tissue integrity yields cellular debris and an increase in extracellular $[K^+]$ and $[H^+]$. Thus, in injured tissues, μM concentrations of H^+ may be found [38]. (iii) Disruption of local capillaries leads to local accumulation of serum products such as complement and products of hemolysis such as serotonin [39,40]. This injury leads to activation of the clotting cascade and kininogens, resulting in the formation of bradykinin [41]. (iv) These products (such as the sP released from the local C fibers) serve to activate mast cells. The mast cells secrete histamine, tryptase, and chymase, which, along with macrophages, secrete a large number of

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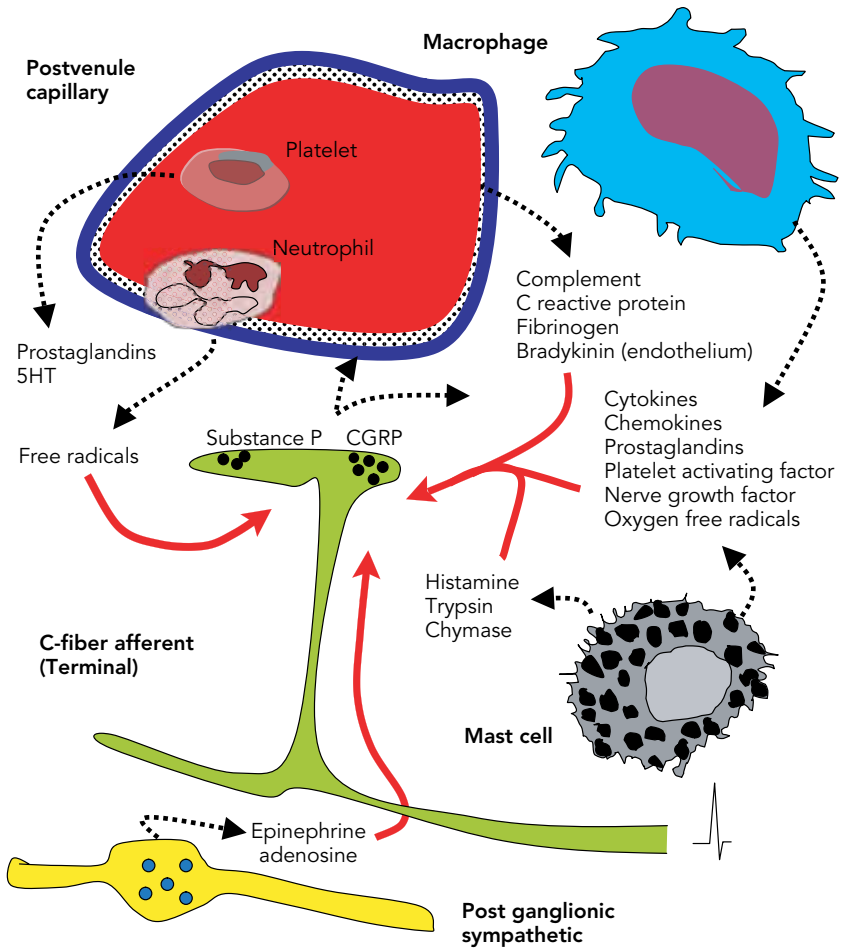


Figure 17.2. Summary of the injury milieu of the primary afferent, including the local vasculature, macrophages, and mast cells and sympathetic terminals and products that are released secondary to that injury which can influence afferent activity. See text for additional details.

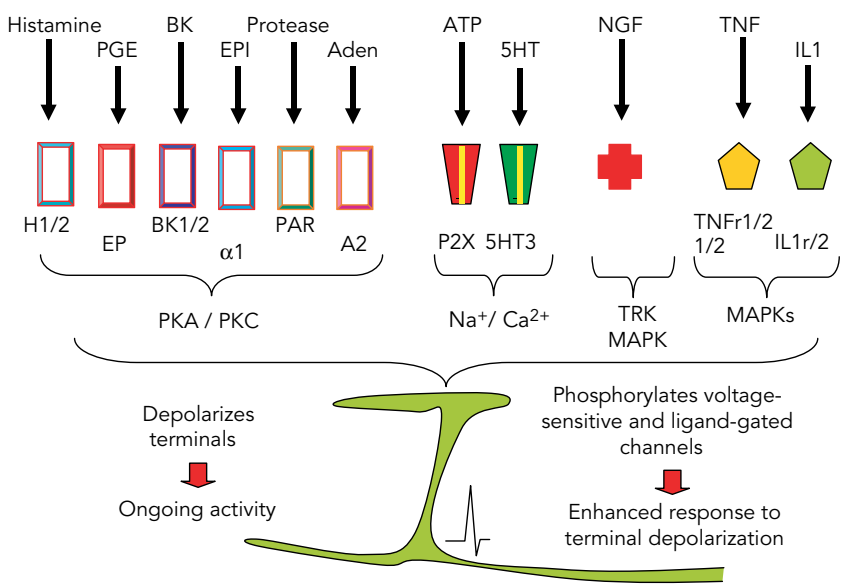


Figure 17.3. Summary of receptor proteins present on the terminals of small primary afferents. Occupancy of the receptor protein by the respective ligand leads to activation of the indicated intracellular messages and/or channel permeability. See text for additional commentary.

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cytokines (IL-1, TNF), chemokines, lipid mediators (prostaglandins, platelet activating factor), and growth factors (NGF) [42,43]. (v) These products also activate integrins, which promote local neutrophil adhesion to the vascular wall and their subsequent diapedesis. The neutrophils migrate in the interstitial fluid along gradients established by a variety of chemotactic stimuli generated by the injured tissue and foreign materials [44]. Neutrophils generate reactive oxygen species and hydrolytic enzymes [45]. These injury events and the antidromic activity on the C-fibers leading to sP and CGRP release are mechanistically responsible for the "triple response of Lewis" noted after tissue injury: a red flush around the site of the stimulus (local arterial dilation), a local edema (increased capillary permeability) and, as will be discussed further later, a local reduction in the magnitude of the stimulus required to elicit a pain response, that is, a hyperalgesia. The chemical complexity of the local injury environment has prompted it to be designated as an "inflammatory soup" [46]. These local cascades associated with tissue injury represent the initiating events associated with the ongoing pain state and the enhanced responsiveness of the afferent axon to a given stimulus.

Our understanding of the cellular and molecular bases of transduction of painful stimuli has burgeoned in the past year, mainly as a result of studies on isolated sensory neurons in culture. The ion channels' underlying neuronal responses to noxious heat, to protons, and to ATP have recently been characterized. The typical increase in nociceptor sensitivity produced by tissue damage has been found to be mediated by at least two distinct mechanisms. In the first, bradykinin and other mediators augments the current activated by heat through a mechanism that involves activation of protein kinase C. In a second sensitization mechanism, prostaglandin E2 for example alters the voltage threshold of several ion channels, including a novel tetrodotoxin-insensitive Na^+ channel, in such a way that initiation of action potentials is facilitated [47,48].

Terminal Receptors for Inflammatory Products

Based on immunohistochemistry as well as physiological studies defining the effect of many of these mediators released by local tissue injury and inflammation, there are eponymous receptors that can be identified on the afferent terminal or in the dorsal root ganglion cells (Figure 17.4). Several examples can be cited.

pH. H^+ ions can directly depolarize afferents [49] through the TRPV1 cation channel [50] and several acid-sensing (ASIC) Na-channels [50]. Thus, TRPV1 channels display a sustained nonselective cationic current at a pH < 6 and are present on a limited population of peptidergic C fibers [51].

Prostaglandins: Agents are synthesized by lipoxygenase (leukotrienes) or cyclooxygenase (prostanoids) upon the release of cell membrane–derived arachidonic acid secondary to the activation of phospholipase A_2. A number of prostanoids, including PGE_2, can directly activate C fibers. Others, such as PGI_2 and TXA_2, and several leukotrienes, can markedly facilitate the excitability of C fibers [52,53]. These effects are also mediated by specific eponymous membrane receptors.

Kinins: Bradykinin released from endothelium can activate two receptors, (constitutively expressed B2 receptors and inducible B1 receptors) to activate free nerve endings of C polymodal nociceptors and generate pain behavior [54]. Bk1/2 receptors act through Gaq and Gai to activate phospholipase A2 (PLA2) and protein kinase C (PKC) indirectly through phospholipase Cb (PLCb) [55]. When given intradermally, Bk produces pain, inflammation and hyperalgesia [56].

Growth factors: Nerve growth factor (NGF) released from fibroblasts, keratinocytes, Schwann cells, and a variety of immune cells can act upon TRK receptors identified in small DRG cells [57]. This action can lead to depolarization of the terminal. SQ NGF can produce a potent sensitization, while anti-NGF strategies can diminish the injury-induced facilitated state. Importantly, over the longer term, NGF serves to regulate expression of the neuropeptides, as well as receptors including TRPV-1 and bradykinin receptors, and voltage-gated sodium channels (see [58]). Glial derived nerve growth factor (GDNF) has been shown to sensitize small afferent terminals in vitro and produce thermal hyperalgesia [59].

Proteinases: Serine proteases are released from mast cells (tryptase), epithelial cells, and/or from the local vasculature. These proteases activate several families of G protein–coupled protein receptors (PAR 1–4) by cleaving a local N-terminal tethered ligand, which binds to and activates receptors present on small afferent terminals [60]. Intradermal injections will activate free nerve endings and produce an enhanced behavioral sensitivity in the animal to subsequent stimulation, for example hyperalgesia [61].

Cytokines: Cytokines such as tumor necrosis factor, interleukin-1 (IL-1), IL-6, and the chemokine IL-8 are released from a variety of inflammatory cells and have been shown to exert powerful sensitizing effects on C-fibers [62,63]. Cytokines appear to induce sensitization via receptor-associated kinases and phosphorylation of ion channels. In chronic inflammation, transcriptional upregulation of receptors has been observed [64]. Intradermal injections of these agents generally produce both mechanical and thermal hyperalgesia, while binding proteins directed against TNF and IL-1β reduce hyperalgesia in inflammatory models [65].

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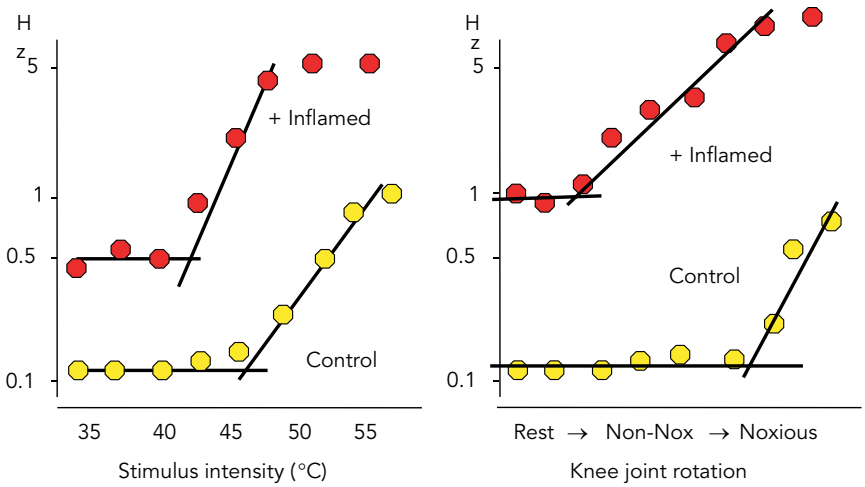


Figure 17.4. Firing of small afferents in the skin (left) occurs with increasing frequency with increasing temperatures. Following carrageenan injection into skin, afferent shows increasing spontaneous activity, a left shift, and an increase in slope of stimulus-response curve, indicating a facilitated response to the thermal stimulus (Right). Firing of articular afferent under normal state and in presence of nonnoxious and then noxious rotation of knee. After injection of irritant into joint, mild rotation results in significant discharge.

Intraterminal Coupling

These peripheral terminal receptors are coupled to a variety of terminal protein kinases (protein kinases A and C as well as mitogen-activated protein kinases, MAPK). These kinases phosphorylate consensus sites found in a variety of transducer proteins (e.g., the TRPV1) and ion channels (e.g., voltage-sensitive sodium channels-NaV) [66]. NGF activation of trkA activation leads to tyrosine phosphorylation of intracellular targets, including ion channels, leading to their enhanced activation by any given degree of terminal activation. These intraterminal cascades represent an important link in the events that lead to terminal sensitization in the face of local injury and inflammation in various organ systems (e.g., skin and joint), such that modest stimuli lead to a pronounced responsiveness [67].

Role of Terminal Sensitization in Pain Behavior

The prominent role of these inflammatory products in the generation of post injury/inflammation pain cannot be overstated. The majority of C-fibers are referred to as "silent nociceptors," as they have little or no spontaneous activity and thresholds so high as to be activated only by extreme physical stimuli. Preclinical studies have emphasized that signs of ongoing pain behavior (paw lifting, paw guarding) is dependent on ongoing activity in these C-fibers [68]. In the presence of the injury products, these terminals can be activated by relatively mild stimuli. Such sensitizing motifs have been described for all tissues including skin [29], bone [69], joint [70], GI mucosa [71], bladder [72], meninges [73], and tooth pulp (see [74]). Accordingly, it can be considered to be an important unifying principle for understanding the sensory components associated with injury. Consistent with these stimulatory and facilitatory effects on afferent function, the local delivery of these products into the skin or viscera have typically been shown to evoke spontaneous pain behavior and a behaviorally defined hyperalgesia. Conversely, local inhibition of the synthesis of many of these factors or their receptors may attenuate the proinflammatory pain state. It is important to note that while any single agent may serve to initiate sensitization of the terminal, it does not necessarily follow, given the complexity of the inflammatory soup, that blocking any single product will significantly attenuate the observed injury/inflammation-induced hyperalgesia.

Time Dependence of Inflammation-Evoked Effects

Injury and inflammation can lead to a relatively acute sensitization (e.g., minutes to hours). However, it should be additionally noted that afferent-mediated downstream effects initiated by inflammation may evolve over time. Thus, inflammation may result in an enhanced expression of a number of terminal channels and receptors that may lead to enhanced sensitivity. Several examples are channels such as voltage-sensitive sodium [75] and calcium channels [76] and afferent neurotransmitters such as sP and CGRP.

The mechanisms by which peripheral inflammation drives changes in terminal transducer protein expression may result from an increase in afferent traffic and DRG depolarization, an effect of local mediators

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to which the DRG may be exposed [77], and/or transport from the periphery to the DRG of a variety of factors from the injury site. Such transported factors may include NGF (nerve growth factor) or cytokines such as TNF released from Schwann cells and inflammatory cells. All of these products may initiate increased DRG transcription and transport of a variety of proteins related to the inflammatory response [78].

CENTRAL COMPONENTS CONTRIBUTING TO INFLAMMATORY PAIN STATES

In the preceding section, the generation of afferent traffic initiated by tissue injury and inflammation was emphasized. This afferent traffic activates dorsal horn neurons. Given the change in afferent transduction in the face of local inflammation, it is reasonable to consider that such peripheral changes may account for the majority of the inflammatory pain phenotype. While important, it is evident that in the face of such peripheral injury, the response of the second-order neuron itself undergoes a significant alteration in responsiveness. Broadly speaking this effect upon second-order excitability reflects events mediated by afferent input as well as (more speculatively) circulating factors that directly effect neuraxial elements (such as microglia).

Afferent Evoked Excitation

Acute input: In the absence of a peripheral stimulus or inflammation, there is minimal if any small afferent traffic. With an adequate stimulus, small afferent input into the spinal dorsal horn is initiated and this input will activate specific populations of second-order neurons, some lying superficially in lamina I (marginal cells) and some lying more deeply in lamina V (wide dynamic range neurons) (Figure 17.5). Many of these neurons that project from the spinal cord to the brain travel in the contralateral-ventrolateral spinal cord (see [28]). The magnitude of the response of the dorsal horn neuron, either wide dynamic range or nociceptive-specific, is related to the frequency (and identity) of the afferent input. As reviewed earlier, the frequency of the afferent input is typically proportional to the magnitude of the acutely applied stimulus.

Persistent input: The organization of this neuraxial response to an acute peripheral stimulus is thus typically modeled in terms of a monotonic relationship between activity in the peripheral afferent generated by the local stimulus and the activity of neurons that project out of the spinal cord to the brain. However, in the face of persistent small afferent input, many dorsal horn neurons will display an enhanced excitability. Intracellular recording has indicated that the facilitated state is represented by a progressive and long-sustained partial depolarization of the cell, rendering

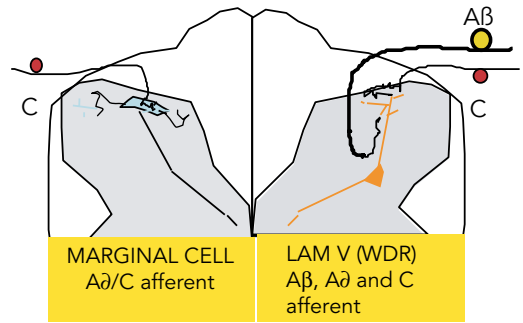


Figure 17.5. Organization of small afferent input into the dorsal horn. Left: small high threshold afferents terminate in the superficial dorsal horn specifically in Rexed Lamina I where they activate marginal lamina I cells. Right: For lamina 5 neurons, there is excitatory input from high threshold afferents that project superficially and from large Aβ axons that terminate deep to the dorsal horn.

the membrane increasingly susceptible to afferent input [79,80]. The pharmacology and organization of this spinal facilitation will be considered further in the following.

Systemic Routes of CNS Activation viz Pain States

In the preceding section, the common thesis was that peripheral injury/inflammation led to the activation of specific populations of sensory afferents innervating the injury site and their input drove a spinal sensitization mediated by a variety of spinal cascades. Such connectivity clearly accounts for the somatotopy associated with local injury, for example, injury to the left hind paw results in a sensitization of the left hind paw mediated in part by the initiation of a facilitated state in the dorsal horn segments receiving input from the left hind paw. An important additional component in the inflammation-induced changes in sensory processing arises from circulating factors. After local trauma or infection, cytokines (such as IL-1β/TNF) are generated and act locally to initiate local inflammatory cascades that can, among their many effects designed to destroy pathogens, serve to sensitize local nerve terminals. In addition, these agents may reach the systemic circulation and generate a system-wide effect, which may be amplified by the subsequent generation of a variety of proinflammatory products such as liver-derived acute phase proteins, pancreatic enzymes, GI products, and granulocytosis [81]. This septic cascade may have potent deleterious effects upon cardiac, pulmonary, and visceral organ function. In addition, these circulating products may have potent effects upon central nervous system (CNS) function, such as temperature regulation (fever), metabolism (cachexia), and pain. The functional linkage between circulating

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inflammatory products and CNS function is not clear. However, ample evidence has pointed to linkages that may involve nonneuronal perivascular cells such as astrocytes and microglia [82,83]. As will be noted later, these non-neuronal cells play an important constitutive role in spinal nociceptive processing after peripheral tissue injury. Such systemic effects could account for bilateral (even whole body) effects. Thus, IV LPS or IL-1β will produce bilateral increases in acute and chronic signs of CNS effects such as bilateral increases in cFOS expression and bilateral increases in the inducible spinal COX-2. Such systemic effects of circulating products may account for reports of bilateral changes after unilateral peripheral inflammation [84].

Functional Consequences of Spinal Sensitization

Sensory transmission: There are three functional consequences of this facilitated spinal state. (i) There is a nonlinear increase in the spinal input–output function, for example, a higher frequency and more persistent discharge for any given afferent input. (ii) Low threshold tactile stimulation also becomes increasingly effective in driving these neurons. (Note that the so-called wide dynamic range Lamina V neurons noted above receive both small and large afferent input.) (iii) In addition to the augmented response of the neuron, the conditioning has the added effect of increasing the receptive field size of the neurons. This occurs because neurons in any given spinal segment receive depolarizing input not only from the spinal root projecting to that segment, but collateral afferent input from adjacent spinal segments which may provide subeffective degrees of depolarization. In the face of sensitization, this subeffective input becomes sufficient to drive depolarization of that neuron. Hence, the effective receptive field of that neuron is increased.

Given the likelihood that the discharge frequency of these dorsal horn neurons contributes to the encoding of a high threshold stimulus as aversive, and that many of these neurons project through the ventrolateral quadrant of the spinal cord (i.e., spinobulbar or spinothalamic projections), this augmented response to a given stimulus and the enlarged receptive field is believed to contribute to the hyperalgesic states observed after nerve injury [28]. Dorsal horn systems that lead to this facilitated state are complex but reflect upon cascades that are in part initiated by the persistent small afferent input and perhaps to a lesser degree by circulating factors.

Inflammation: The sensitization of the spinal systems leads to an enhanced pain response. What is less appreciated is that sensory afferents conduct both orthromically (from the outside to the spinal cord) and antidromically (from the spinal cord to the outside). As will be reviewed later, the peripheral terminal of the primary afferent is able to release a variety of products such as peptide transmitters (substance P/CGRP). Many of these products are believed to control the local inflammatory response. It is appreciated that central sensitization can enhance the excitability of the central terminal leading to primary afferent depolarization (PAD) and significant amounts of antidromic traffic [85,86]. Although there is not enough space to discuss the ramifications of this phenomenon, there is ample data to suggest that such antidromic linkages may additionally contribute to the local inflammatory response and sensitization [87,88].

COMPONENTS OF THE DORSAL HORN SYSTEMS CONTRIBUTING TO SPINAL SENSITIZATION

As noted earlier, persistent small afferent input as generated by small afferent input initiated by tissue injury and inflammation will lead to a sensitized behavioral state. This sensitization results from a series of cascades, which reflect the organization of dorsal horn systems that regulate the activity in neurons (such as the Lam I – marginal neurons and Lam 5 – wide dynamic range neurons) that project to higher centers. In the following narrative, representative examples of the contribution played by these several systems will be reviewed [89].

Local Neuronal Excitability

Afferent evoked membrane depolarization: As indicated in Figure 17.6, small afferents contain and release into the spinal dorsal horn excitatory amino acids such as glutamate and peptides such as substance P. Acute activation yields second-order neuronal depolarization mediated largely by glutamate acting through an AMPA receptor Na^+ ion-selective channel, which produces a potent but transient depolarization of the membrane [90,91]. As reviewed earlier, repetitive small afferent input will yield a facilitated state in the dorsal horn as exemplified by the phenomena of "wind up." The pharmacology of this central facilitation suggests that the spinal phenomenon reflects more than just the repetitive activation of a simple excitatory system. Thus, while the acute excitation is unaffected by N-methyl-D-aspartate (NMDA) receptor antagonists, these agents will prevent spinal wind-up [92]. In parallel with these physiological observations, behavioral work demonstrated that the spinal delivery of such drugs will have no effect upon acute pain behavior, but reduce the facilitated states induced after tissue injury [93]. This differential effect upon persistent versus acute activity results from the fact that the NMDA ionophore under resting membrane potential is blocked by a Mg^{2+} ion in the channel. This block is removed in the face of persistent membrane depolarization, such as that which occurs with repetitive

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afferent input. Activation of the NMDA ionophore leads to a marked increase in intracellular Ca^{2+} that initiates a variety of downstream facilitatory events. A variant of the AMPA channel occurs in a small proportion of dorsal horn channels and creates an AMPA-calcium permeable ionophore that can also contribute to downstream facilitation [94]. In addition to the classic transmitters, growth factors such as brain-derived nerve growth factor (BDNF) is synthesized by small DRGs and released from spinal terminals, packaged in dense-cored vesicles, and transported within axons into terminals in the dorsal horn of the spinal cord. BDNF has a potent sensitizing effect on spinal neurons [95] (see Figure 17.6).

Activation of kinases: Persistent afferent input leads to a marked increase in intracellular Ca^{2+} through several mechanisms: (i) voltage-gated ion channels (e.g., N and T type channels), (ii) ion-gated channels (e.g., NMDA/calcium-permeable AMPA channels), and (iii) G-protein–coupled receptors that act to mobilize intracellular calcium (e.g., neurokinin-1 receptors). The increased calcium leads in turn to the activation of a wide variety of phosphorylating enzymes, including protein kinases A and C, calcium calmodulin-dependent protein kinases, as well as mitogen-activated kinases (MAPKs) including p38 MAP kinase and ERK (see [66]), which increases the excitability of a variety of spinal linkages. Several examples of such local intracellular cascades will be noted.

(i) The functionality of many channels is regulated by phosphorylation. One such channel is the NMDA receptor. Such phosphorylation of spinal NMDA receptors leads to increases, and dephosphorylation leads to decreases in NMDA currents [96]. These events initiated by repetitive afferent activation serves to reduce the threshold for activation, leading to an enhanced response of the NMDA ionophore to further depolarization. Block of spinal PKA and PKC activity can reduce behaviorally defined hyperalgesia otherwise observed after tissue injury [97].

(ii) P38 MAPK is activated (phosphorylated) by an increase in intracellular Ca. This activation leads to several events. The first is to phosphorylate phospholipase A2 (PLA2), which initiates the release of arachidonic acid and provides the substrate for cyclooxygenase (COX) to synthesize prostaglandins. The second is that this MAPK activates a variety of transcription factors (such as NF-κB), which activates the synthesis of a variety of proteins, including COX-2. The spinal delivery of P38 MAPK inhibitors will thus reduce acutely initiated hyperalgesia and reduce the upregulation of COX-2 otherwise produced by injury [98,99].

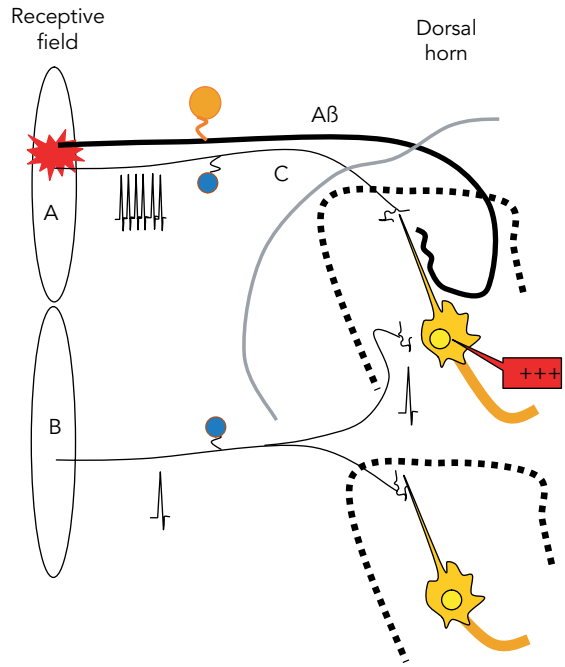


Figure 17.6. Injury to receptive field (RF) A leads to a persistent afferent barrage which strongly excites and sensitizes the second order neuron. This sensitization has three consequences to the response properties of Neuron A. (i) Neuron A gives a greater response for any given stimulus. (ii) This cell (a WDR neuron) receives large low threshold (Aß) input and that input now evokes a greater response. (iii) Collateral input from receptive field B now is sufficient to activate effects to activate the sensitized neuron A. This means that the effective receptive field of neuron A is now enlarged to include RF A and RF B, for example, stimuli now applied to RF-A and B will activate Neuron A. This connectivity likely accounts for the enlarged receptive field observed after peripheral injury, for example, 2° hyperalgesia.

Lipid cascades: A variety of phospholipases (c, i, and s-PLA2) and cyclooxygnease (COX-1 and 2) are constitutively expressed in the dorsal horn in both neuronal and nonneuronal cells) [100]. Not surprisingly, downstream lipid products such as prostaglandins are synthesized and released into the spinal dorsal horn and this release can be markedly enhanced by small afferent input generated by tissue injury and by the direct activation of second-order neurons [101]. These agents diffuse extracellularly to act presynaptically to enhance the opening of voltage-sensitive calcium channels, thus augmenting transmitter release [102]. In addition, prostaglandins act postsynaptically to *block* glycinergic inhibition on second-order dorsal horn neurons [103]. Such a reduction in the activation of inhibitory glycine or GABA interneuron regulation can lead to a potent facilitation of dorsal horn excitability (see later). The spinal delivery of PGE will increase, while PLA2 or COX-2 inhibitors will reduce spinal PGE2 release and reduce injury-induced hyperalgesia.

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Nitric oxide synthase (NOS): NOS forms diffusible nitric oxide (NO) from arginine. There are three principal NOS isoforms: endothelial, neuronal, and inducible. The neuronal and inducible forms have been found to play a facilitatory role in the spinal cord through the formation of NO, which acts presynaptically through cGMP to enhance transmitter release. Spinal NOS inhibitors can reduce posttissue injury hyperalgesia [104,105].

Modulation of Tonic Inhibitory Circuits

Second-order dorsal horn neurons (Lamina V WDR neurons) receive robust excitatory input from large (Aß) afferents. In spite of this afferent input onto dorsal horn neurons that are believed to play a role in nociceptive processing, such large afferent input does not lead normally to a pain state (Figure 17.7). As reviewed earlier, after local tissue injury such tactile stimulation may lead to a sensitization to light touch (tactile allodynia). An important component of this sensitization is believed to be related to a loss of local GABAergic or glycinergic inhibition. Thus, dorsal horn GABAergic and glycinergic interneurons receive excitatory drive from low-threshold afferents and in turn form synapses pre- and postsynaptic to the large primary afferent. Accordingly, the postsynaptic excitation by large afferent input is modulated by the concurrent activation of these inhibitory interneuons. The blocking of spinal GABA A and glycine receptors by spinal receptor blockers will lead to a marked exaggeration of the excitation evoked by Aß input and will induce a behavioral phenotype in which low-threshold tactile stimulation will induce a prominent pain state [106,107]. Three circumstances may alter this GABAergic/glycinergic modulation. (i) Loss of GABA and glycine neurons or a downregulation in the respective receptors. While early work did suggest such changes may occur after nerve injury, such changes have not been demonstrated to be robust and are unlikely to occur after inflammation [108]. (ii) As reviewed earlier, small afferent input will lead to a release of a number of lipidic acids including PGE_2. PGE_2 will reduce the glycine-mediated opening of the glycine ionophore, leading to a reduction in this local modulation [103]. (iii). The GABA A and glycine receptors are Cl^- ionophores. At the postsynaptic membrane increasing Cl^- conductance typically leads to a mild hyperpolarization of the membrane. It is now appreciated that in the face of peripheral tissue (or nerve injury) there is an alteration in the relative Cl^- exporter/importer activity leading to an increase in the Cl^- equilibrium potential and an enhanced depolarization with increased Cl^- permeability [109]. These events would lead to the enhanced response to low-threshold afferent input observed after peripheral tissue injury and inflammation.

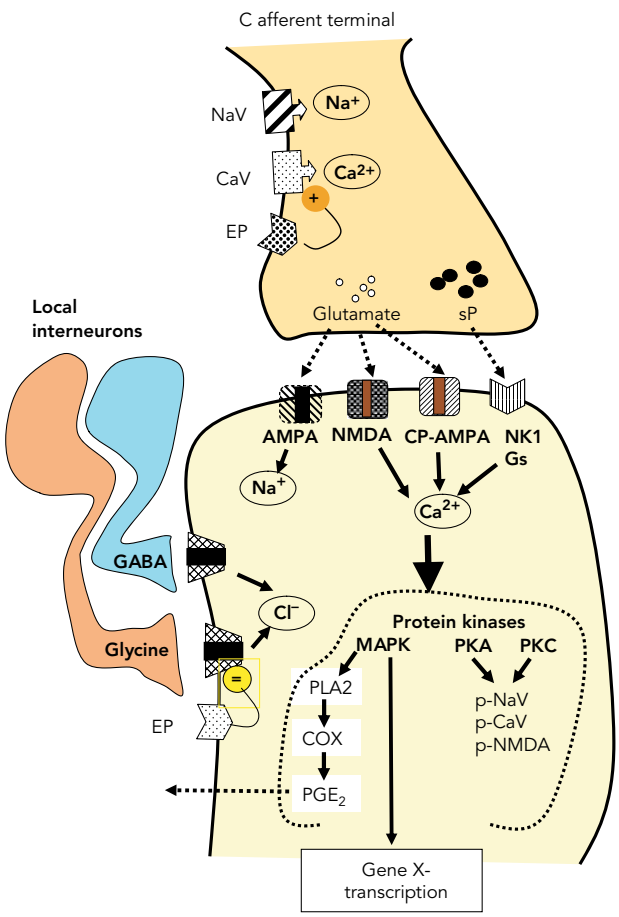


Figure 17.7. First-order synapse for a small afferent axon in the spinal dorsal horn. As indicated, there is the release of small afferent transmitters which depolarize the second order neurons through glutamate (AMPA/NMDA0 and sP (Neurokinin1: NK1) receptors). These serve to depolarize the membrane by increasing cation conductance. This in turn activates downstream cascades that serve to sensitize the membrane to additional depolarization through a variety of mechanisms. See text for additional commentary.

Bulbospinal Systems

Serotonergic pathways (originating in the midline raphe) and noradrenergic pathways (originating in several brain nuclei including the locus corueleus) project into the spinal dorsal horn (see Figure 17.8). Serotonergic systems act through a variety of dorsal horn receptors that may be either inhibitory (5HT1a/b) or directly excitatory (5HT2/3). As indicated schematically in Figure 17.8, small afferent input will activate superficial Lamina I dorsal horn neurons that project into the medulla to activate raphe spinal projections that excite deep dorsal horn (lamina 5) neurons by 5HT3 receptors. Blocking of this circuit has been shown to prominently reduce dorsal horn facilitatory conditioning by small afferent input [110].

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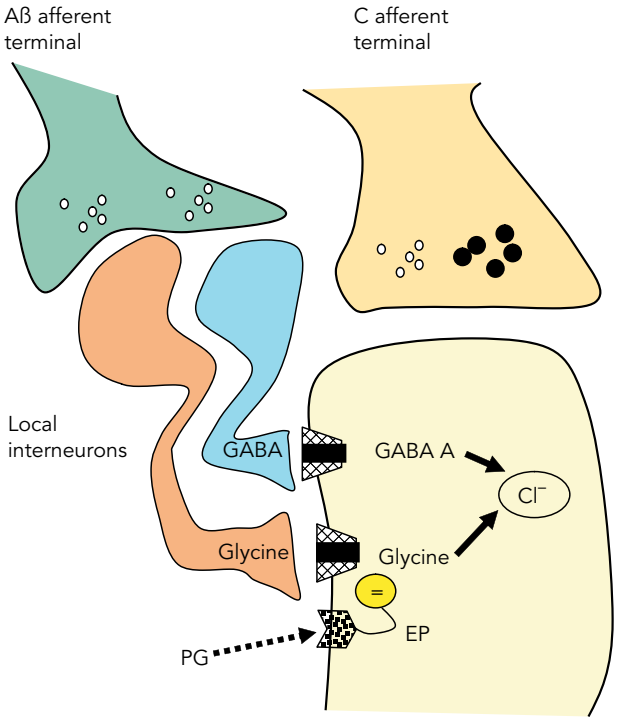


Figure 17.8. Small afferent input activates dorsal horn nociceptive neurons. Large afferent input also activates nociceptors, but as indicated these large low-threshold primary afferents additionally activate inhibitor interneurons (GABA and glycine) which serve to attenuate the excitatory drive produced by large afferent input. Though the interneurons are shown on the postsynaptic membrane they may also be presynaptic on the large myelinated afferent. The outcome is to diminish the excitation otherwise produced by the large afferents. Loss of that inhibition would lead to a prominent increase in large afferent-evoked excitation.

Non-Neuronal Cells

In the CNS, there are a variety of nonneuronal cells. Among these are astrocytes and microglia. Microglia are resident macrophages that appear in the brain from the circulation during development. While classically these non-neuronal cells have been shown to play important roles in a variety of trophic functions such as blood-brain barrier integrity, extracellular water balance, glucose transport, and phagocytic functions in the face of injury, current thinking has begun to emphasize their contributions to the excitability of local neuronal circuits. These astrocytes, microglia and neurons form a complex network in which each can acutely influence the excitability of synaptic transmission [111]. Thus, non-neuronal cells can influence synaptic transmission by the release of a wide variety of proexcitatory products (such as ATP, superoxide/free radicals, nitric oxide, and cytokines). Glial cells can also regulate extracellular parenchymal glutamate by their glutamate transporters. This can serve to increase extracellular glutamate activating neuronal glutamate receptors [112] (see Figure 17.9). These non-neuronal networks can be activated by transmitters from primary afferents, and intrinsic neurons releasing glutamate, sP and ATP can overflow from the synaptic cleft to these adjacent non-neuronal cells and lead through eponymous receptors to their activation. Astrocytes can form local networks which may communicate over a distance by the spread of excitation through local nonsynaptic contacts ("gap" junctions) [113]. Further, astrocytes may communicate with other astrocytes and microglia by releasing a number of products including glutamate/cytokines and "S100" protein [114]. Neurons may activate microglia by the specific release of a membrane chemokine (fractalkine), which acts on specific receptors found on microglia. In addition to afferent input after tissue injury and inflammation, *circulating* cytokines (such as IL-1β/TNF can activate perivascular astrocytes/microglia. As noted, microglia are in fact brain resident macrophages. Many circulating products referred to as "acute phase reactants" (TNF, prostaglandins (PGE_2), sympathetic amines and leukotriene-B_4) can initiate sensitized pain states and may account for mechanical hypersensitivity which arises, for example, in allergenic reactions in rat models [64,115].

Importantly, the role of these non-neuronal cells in inflammation and injury-induced pain states is supported by the observations that spinal inhibitors of microglial activation, such as minocycline (a second generation tetracycline) and pentoxyfilline, have been reported to block indices of acute and chronic glial activation and diminish hyperalgesic states [82]. Similar metabolic inhibitors that block astrocyte activation (flurocitrate) can similarly diminish hyperalgesia after nerve and tissue injury. These agents, while not clinically useful, suggest important directions in drug therapy development.

These processes outlined earlier represent a complex cascade referred to broadly as "neuroinflammation" (Figure 17.9). Importantly, while indices of glial activation (e.g., increased expression of epitopes such as GFAP and OX42) are considered hallmarks of neuroinflammation as observed after local injury, current thinking emphasizes that astrocytes and microglia are indeed *constitutively* active and contributing to acute changes in spinal network excitability [116].

In conclusion, tissue injury and inflammation lead to a surprisingly complex behavioral phenotype characterized by ongoing pain and the appearance of states wherein moderate noxious or frankly innocuous stimuli will lead to an augmented pain state at the injury site (primary) and adjacent to the injury site (secondary). The mechanisms underlying these behavioral states are equally complex, reflecting the release of active factors at the injury site, which initiates afferent traffic and produces a sensitization of the afferent terminal leading to an enhanced response to a given stimulus. In addition, the ongoing activity initiated

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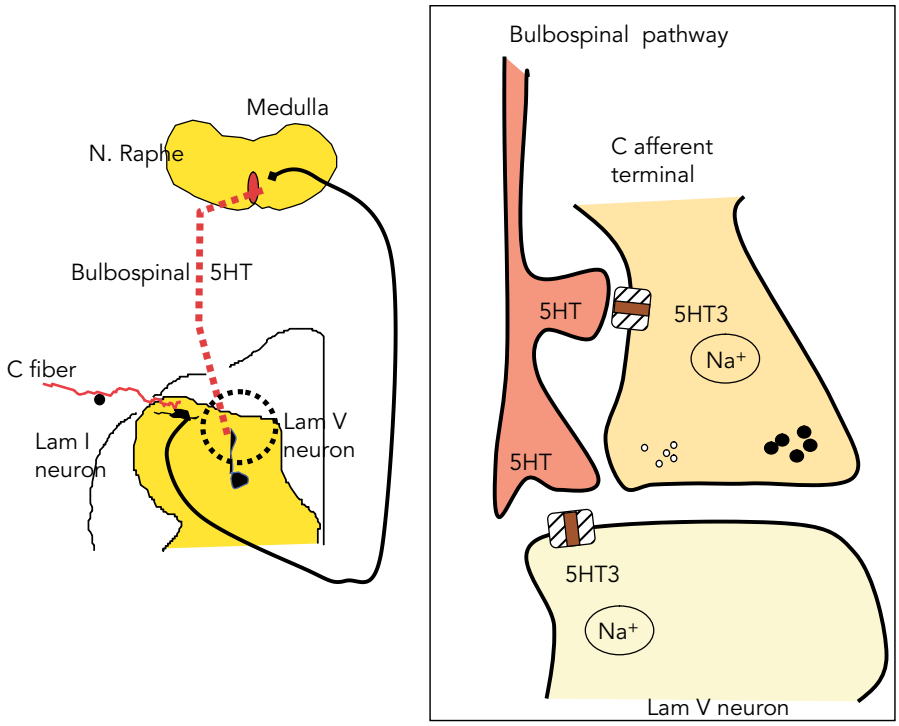


Figure 17.9. Small afferent input activates ascending projections for Lamina V neurons that activate raphe spinal neurons that send serotonin projections to the spinal cord. At the spinal level 5HT released from the bulbospinal projection activates the second-order lamina V neuron through a $5HT_3$ receptor.

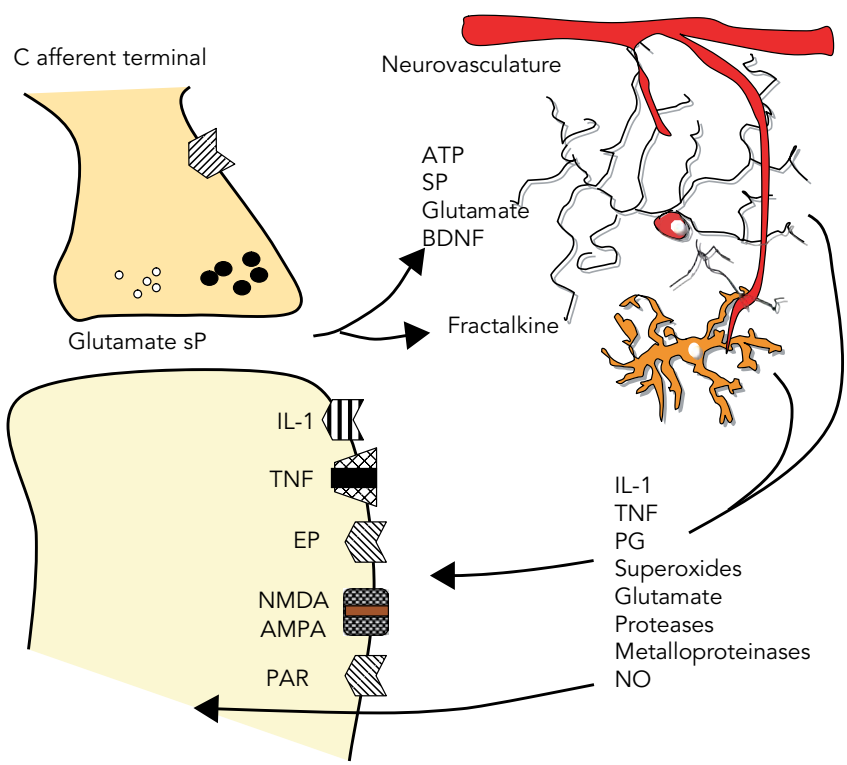


Figure 17.10. Dorsal horn non-neuronal cells (microglia/astrocytes) are activated by overflow from dorsal horn synapses, from second-order neuron, from each other, and through products in the cerebral vasculature. Such activation leads to the secretion of a large variety of inflammatory products that can influence the excitability of neuronal transmission.

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by injury leads to a powerful spinal facilitation of spinal sensory networks that is mediated by increases in local circuit excitability, activation of non-neuronal cells and extraspinal feedback loops (Figure 17.10). Interestingly, the role of spinal non-neuronal cells provides an important link between circulating factors generated by the body's response to injury and inflammation and the central nervous system processing. It is of particular interest in that a variety of Toll receptors are present on both microglia and astrocytes. These recptors likely mediate the efects of circulating products (such as LPS) on CNS function and it is increasingly likely that products released within the neuraxis may also serve to modulate the role played by these non neuronal cells in regulating spinal excitability in the face of persistent afferent input and tissue injury. Importantly, these scenarios find their expression in *all* innervated systems, from the skin to the inner lumen of visceral organs. Conversely, as reviewed, antidromic activity on sensory afferents can modulate the peripheral inflammatory response.

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18 Nonsteroidal Anti-Inflammatory Drugs

Samir S. Ayoub and Roderick Flower

INTRODUCTION

Nonsteroidal anti-inflammatory drugs (NSAIDs), sometimes called the "*aspirin-like drugs*," are among the most widely used of all drugs. Aspirin itself was introduced by Bayer in 1898 as a replacement for salicylic acid, which had been available in synthetic form since the 1870s and as the active constituent of plant (e.g., willow bark) preparations for many centuries before that. Since the beginning of the 20th century, the number of NSAIDs has grown substantially: phenylbutazone was introduced in the 1940s, the fenamates in the 1950s, indomethacin in the 1960s, the proprionates in the 1970s, and the oxicams in the 1980s. The 1990s saw a radical new development – the introduction of "coxibs" and the new millennium has ushered in an era of reappraisal and reassessment of our understanding of the pharmacology of these drugs and their therapeutic and unwanted effects. There are now more than 50 different NSAIDs on the global market and some of the more prominent examples are listed in Figure 18.1. (Chemical structures of NSAIDs shown, grouped according to chemical structure.)

NSAIDs had been in clinical use for a long time prior to an understanding of their pharmacological mechanism of action. The real breakthrough in NSAID pharmacology was made in the early 1970s when John Vane identified that NSAIDs work by inhibition of prostaglandin (PG)-forming cyclo-oxygenase (COX) activity resulting in the reduction of prostanoid synthesis. The second breakthrough came about in the early 1990s with the discovery of COX-2 that eventually lead to the development of the COX-2-selective NSAIDs that lacked some of the major side effects of classical NSAIDs.

Clinically, the NSAIDs provide symptomatic relief from pain and swelling in chronic joint disease such as occurs in osteo- and rheumatoid arthritis as well as in more acute inflammatory conditions such as sports injuries, fractures, sprains, and other soft tissue injuries; can also alleviate postoperative, dental, and menstrual pain, as well as headaches and migraine. Many NSAIDs are available in a variety of different formulations such as tablets, injections, and gels and several NSAIDs are available from pharmacies "over-the-counter" without prescription. The latter are taken in large quantities for many types of minor aches and pains – and occasionally abused. Aspirin itself is still consumed in prodigious amounts around the world and new uses are continually being found for this drug. Virtually all NSAIDs, particularly the "classical" NSAIDs, can have significant unwanted effects, especially in the elderly. Newer agents have less severe adverse actions.

PRINCIPAL PHARMACOLOGICAL ACTIONS

NSAIDs are sometimes known as the aspirin-like drugs because their pharmacology is broadly similar to that of aspirin the "archetypal" NSAID. Their three main therapeutic effects are an *anti-inflammatory* effect – reducing the symptoms of inflammation; an *analgesic* effect – reduction of certain types of (especially inflammatory) pain and an *antipyretic* effect – lowering the body temperature when this is raised in disease (i.e., fever). It is important to note that whilst NSAIDs relieve many of the symptoms of inflammation, they have little or no action on the actual progress of underlying chronic diseases itself. As a class, they are generally without effect on other aspects of inflammation such as leukocyte migration, T-cell function, lysosomal enzyme release, toxic oxygen radical production, and so on that contribute to tissue damage in chronic inflammatory conditions such as rheumatoid arthritis, vasculitis, and nephritis.

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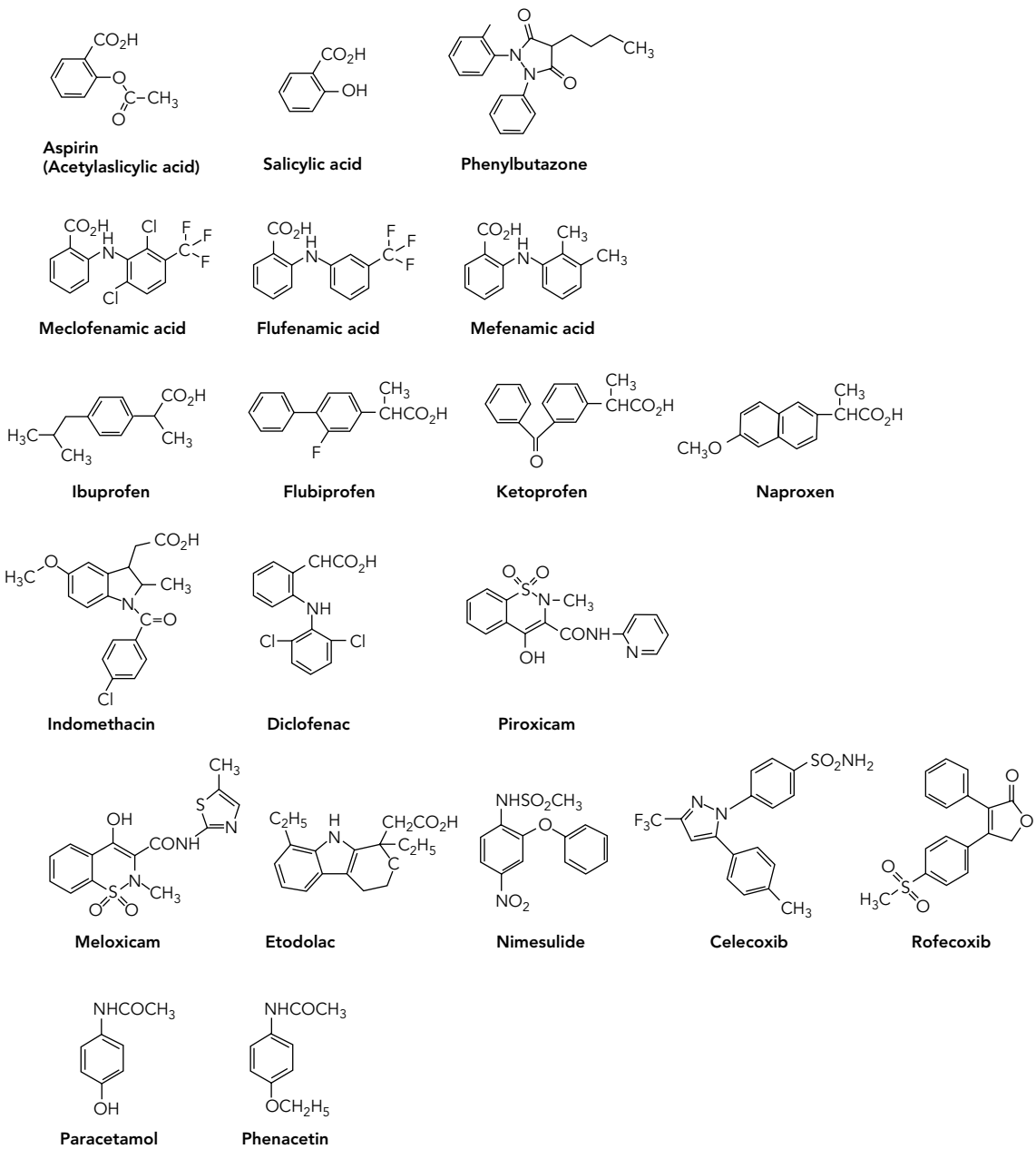


Figure 18.1. The chemical structures of some NSAIDs and "coxibs" representative of the main classes of these drugs. Note that whilst NSAIDs are often carboxylic acids, the coxibs are not. This has some relevance to their isoform selectivity (see Figure 18.2).

We Will Review Each of Their Principal Actions in Turn

Anti-Inflammatory Effects

Many mediators act to initiate and coordinate acute inflammatory and allergic reactions. Whilst some are produced in response to specific stimuli (e.g., histamine in allergic inflammation) there is considerable redundancy and each facet of the response – vasodilatation, increased vascular permeability, cell accumulation, and so forth – can generally be produced by several separate mechanisms.

NSAIDs reduce mainly those components of the inflammatory and immune response in which prostaglandins, predominantly derived from COX-2 (see later), play a significant part. These include *vasodilatation*, in which the blood flow to inflamed tissues is increased by a direct vasodilator action of prostanoids (giving rise to the "redness" of inflammation); and *oedema*, this is brought about through an indirect potentiation of the effect of other agents

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(e.g., histamine) which directly increase the permeability of postcapillary venules leading to increased fluid movement through the microvasculature and giving rise to the "swelling" of inflammation.

In experimental animals, NSAIDs have been shown to prevent resolution of the inflammatory reaction if administered toward the end of the acute inflammatory process. These findings provide supportive evidence for an anti-inflammatory role for some prostaglandins, which have been shown to increase at the latter phase of acute inflammation.

Analgesic Effects

The NSAIDs are suitable for the treatment of many types of mild-moderate pain including inflammatory and postoperative pain. They are generally without action in neuropathic pain. NSAIDs provide a different quality of analgesia to that of the opiates in the sense that they do not affect other sensory modalities. Their analgesic "ceiling" is also different. Two sites of action have been identified: peripherally, NSAIDs decrease production of the prostaglandins that sensitize nociceptors to the direct action of inflammatory mediators such as bradykinin. In doing so they relieve the hyperalgesia in conditions that are associated with increased local prostaglandin synthesis such as arthritis, bursitis, pain of muscular and vascular origin, toothache, dysmenorrhoea, the pain of postpartum states, and the pain of cancer metastases in bone. They may be administered in combination with opioids and in some cases can substantially reduce the requirement for opioids required to produce an analgesic effect. Their ability to relieve some types of headache (less frequently migraine) may be related to the abrogation of the vasodilator effect of prostaglandins on the cerebral vasculature.

In addition to these peripheral effects, there is a less well-characterized central action possibly in the spinal cord. Inflammation increases COX-2 expression in the cord and local prostaglandin release facilitates transmission by afferent nociceptive fibers, by enhancing the release of excitatory mediators such as substance P and glutamate, to relay neurons in the dorsal horn. Prostaglandins have an inhibitory effect on descending noradrenergic and glycinergic inhibitory pathways that terminate in the spinal cord, thus causing dis-inhibition of spinal nociceptive pathways enhancing facilitation further. Prostaglandins may also activate microglia in the cord that destroy inhibitory interneurones leading to a remodeling and permanent activation of pain pathways.

Other sites of action have been mooted: NSAIDs and COX-2-selective inhibitors injected directly to the preoptic area of the hypothalamus have been shown to induce analgesia in response to systemically administered noxious agents.

Antipyretic Effect

Body temperature is regulated by the preoptic area of the hypothalamus, which functions in a way similar to a thermostat by controlling the balance between heat loss and heat production, thereby regulating the overall temperature of the organism around a "set point" (e.g., normal body temperature). Fever occurs when there is a disturbance of this homeostatic system, which leads to the set-point of body temperature being raised. During an inflammatory reaction, bacterial endotoxins, viruses, or other foreign substances acting through TOLL receptors, cause the release from macrophages and other cells of cytokines such as TNF and IL-1. These substances can directly stimulate the generation, in the hypothalamus, of E-type prostaglandins which elevate the temperature set-point. COX-2 is induced by IL-1 in vascular endothelium in the hypothalamus. NSAIDs exert their antipyretic action largely through inhibition of prostaglandin production in (or close to) the hypothalamus. There is some evidence that prostaglandins are not the only mediators of fever; hence, NSAIDs may have an additional antipyretic effect through mechanisms as yet unknown. Once there has been a return to the normal hypothalamic set-point, the temperature-regulating mechanisms (dilatation of superficial blood vessels, sweating, etc.) then operate to reduce temperature to normal. Normal body temperature in humans is not affected by NSAIDs.

UNWANTED EFFECTS

The NSAIDs also share, to a greater or lesser degree, the same types of mechanism-based side effects although there may be other additional unwanted effects peculiar to individual members of the group. The main effects include *gastric irritation*, which may range from simple discomfort to ulcer formation; an effect on *renal blood flow*, particularly in the compromised kidney; and a tendency to prolong bleeding through inhibition of *platelet function* and *skin reactions*. More controversially, it is argued that all NSAIDs – but especially COX-2 selective drugs – increase the likelihood of thrombotic events such as myocardial infarction, by inhibiting PGI_2 (Prostacyclin) synthesis.

Overall, the burden of unwanted NSAID side effects is high. When used in joint diseases (necessitating fairly large doses and long-continued use), there is a high incidence of side effects – not only in the gastrointestinal (GI) tract but also in liver, kidney, spleen,

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blood, and bone marrow. COX-2-selective drugs generally have less GI toxicity (see later).

We Will Again Look at Each Area in Turn

Gastrointestinal Disturbances

Adverse GI events are the commonest unwanted effects of the NSAIDs and are believed to result mainly from inhibition of the gastric COX-1 responsible for the synthesis of the prostaglandins that normally exert an inhibitory tone on acid secretion and protect the mucosa. Common GI side effects include gastric discomfort, dyspepsia, diarrhea (but sometimes constipation), nausea and vomiting and, in some cases, gastric bleeding and frank ulceration. It has been estimated by some that 34%–46% of users of NSAIDs will sustain some GI damage that, whilst it may be asymptomatic, carries a risk of serious hemorrhage and/or perforation. Severe GI effects alone (perforations, ulcers, or bleeding) are said to result in the hospitalization of over 100,000 people per year in the United States. Some 15% of these patients may die from this iatrogenic disease. These figures probably reflect the fact that NSAIDs are used extensively in the elderly, and often for extended periods of time. The mechanism is dependent on inhibition of COX in the gastric mucosa and damage is seen whether the drugs are given orally or systemically. However, in some cases (aspirin being a good example) *local* damage to the gastric mucosa caused directly by the drug itself may compound the damage. Oral administration of prostaglandin analogues such as misoprostol can diminish the gastric damage produced by these agents and NSAIDs are also frequently given together with H_2 blockers or proton pump inhibitors for the same reason.

On the basis of extensive experimental evidence, it had been predicted that COX-2-selective agents would provide good anti-inflammatory and analgesic actions with less gastric damage and some older drugs (e.g., meloxicam) that were believed to be better tolerated in the clinic, turned out to have some COX-2 selectivity. Two large prospective studies compared celecoxib and rofecoxib with standard comparator NSAIDs in patients with arthritis and showed some benefit, although the results were not as clear-cut as had been hoped.

Skin Reactions

Rashes are common idiosyncratic unwanted effects of NSAIDs, particularly with mefenamic acid (10%–15% frequency) and sulindac (5%–10% frequency). The underlying mechanism is uncertain. Symptoms range from mild erythematous, urticarial, and photosensitivity reactions to more serious and (rarely) potentially fatal diseases including Stevens–Johnson syndrome.

Haemostatic System

All NSAIDs (except COX-2 inhibitors) prevent platelet aggregation because COX-1 is important in generating proaggregatory TxA_2 (thromboxane A_2) and therefore may prolong bleeding. Again, aspirin is the main problem in this regard although this action is also the basis of aspirin's cardiovascular utility (see later).

Renal Effects

Therapeutic doses of NSAIDs in healthy individuals pose little threat to kidney function, but in susceptible patients they cause acute renal insufficiency, which is reversible on stopping the drug. This occurs through the inhibition of the biosynthesis of those prostanoids (PGE_2 and PGI_2) involved in the maintenance of renal blood flow, specifically in the PGE_2-mediated compensatory vasodilatation that occurs in response to the action of noradrenaline or angiotensin II. Neonates and the elderly are especially at risk as are patients with heart, liver, or kidney disease, or a reduced circulating blood volume.

It seems that adverse cardiovascular events especially during prolonged use or in high cardiovascular risk patients may be a feature of all drugs that inhibit COX activity. Hypertension, caused by inhibition of renal COX-2, seems to be a strong possibility. The notion that COX-2 inhibitors selectively increase cardiovascular risk through a selective action on vascular COX-2, thereby diminishing PGI_2 production (the "Fitzgerald hypothesis"), seems less likely.

Chronic NSAID consumption, especially NSAID "abuse," can cause *analgesic nephropathy* characterized by chronic nephritis and renal papillary necrosis. Historically, phenacetin, now withdrawn, was the main culprit; paracetamol, one of its major metabolites, is much less toxic. Regular use of prescribed doses of NSAIDs is less hazardous for the kidney in this respect than very heavy and prolonged use of over-the-counter analgesics.

Other Unwanted Effects

Other, much less common, unwanted effects of NSAIDs include CNS effects, bone marrow disturbances, and liver disorders, the latter being more likely if renal impairment is already present. Paracetamol overdose may cause liver failure if not treated promptly. Approximately 5% of patients exposed to NSAIDs may experience *aspirin sensitive asthma*. The exact

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mechanism is unknown but inhibition of COX is implicated. Aspirin is the worst offender but there is cross-reaction with all other class members except, possibly, COX-2 inhibitors.

MECHANISM OF PHARMACOLOGICAL ACTION

As a group, the NSAIDs are structurally diverse, although most are carboxylic acids (Figure 18.1). The question that went unanswered for almost seven decades was how the apparently disparate therapeutic and side effects were mechanistically linked.

There were several early suggestions, but writing about his breakthrough discovery in 1971, Vane tells us that the idea that the aspirin-like drugs blocked the conversion of substrate arachidonic acid to prostaglandins came to him while reviewing experiments in which aspirin blocked the release of "rabbit aorta contracting substance" (RCS) from guinea-pig and dog lung. Believing that RCS (later shown to be a mixture of TxA_2 and prostaglandin endoperoxides) was an intermediate in prostaglandin synthesis, he wrote *"a logical corollary was that aspirin might well be blocking the synthesis of prostaglandins."*

A quartet of papers appeared that year from Vane and members of his group, showing that aspirin itself, indomethacin and (less effectively) sodium salicylate blocked prostaglandin synthesis in a crude cell-free COX preparation and in isolated perfused spleen of dogs. In humans, therapeutic doses of aspirin taken by volunteers reduced prostaglandin generation by aggregating platelets ex vivo or in seminal plasma samples collected during the course of the treatments. The overall message was clear – at least some NSAIDs were able to prevent the generation of prostaglandins by direct action on the COX enzyme, and did so in humans in clinical doses. But how was this linked to their therapeutic actions?

The late 1960s and early 1970s had seen an explosion of interest in prostaglandin biology. It was observed that prostaglandins were generated during platelet aggregation, and that they produced fever, hyperalgesia, and inflammation. Prostaglandins had also been detected in the gastric mucosa and been shown to inhibit ulcer formation in rodent models of gastric damage and cytoprotection. In other words, the ability of NSAIDs to block COX provided the much sought after link between the therapeutic and side effects of these drugs.

Over the next couple of years, it was shown that the entire gamut of NSAIDs inhibited COX at concentrations well within their therapeutic plasma range and that the overall order of potency corresponded with their therapeutic activity. Other types of anti-inflammatories, such as the glucocorticoids and the so-called disease modifying drugs such as gold and penicillamine, were inactive in these cell-free assays, providing further evidence for the specificity of the NSAID effect. Using active/inactive enantiomeric pairs of NSAIDs, such as naproxen, an exquisite correlation was observed between the anti-inflammatory and anti-COX activity, and many more studies confirmed the notion that this is a fundamental mechanism of this class of drugs.

Other actions besides inhibition of COX may contribute to the anti-inflammatory effects of some NSAIDs. Reactive oxygen radicals produced by neutrophils and macrophages are implicated in tissue damage in some conditions, and some NSAIDs (e.g., sulindac) have oxygen-radical-scavenging effects as well as COX-inhibitory activity so may decrease tissue damage. Aspirin also inhibits expression of the transcription factor NF-κB, which has a key role in the transcription of the genes for inflammatory mediators. A direct toxic action of NSAIDs on the gut may be at least in part mediated through an "uncoupling" effect on epithelial mitochondria.

By 1974, Vane's concept was firmly established as the most powerful explanatory idea in the field. One effect of this was that the pharmaceutical industry now possessed, probably for the first time, a simple and robust in vitro screen for putative anti-inflammatory compounds. Probably as a result of this, the number of chemical abstracts dealing with potential inhibitors of the COX enzyme rose markedly, with more than 2,500 per year recorded within a decade of these ideas taking hold.

However, several anomalies were noted by early workers in the field. For example, paracetamol (acetaminophen) – another hugely popu1ar drug that was also introduced in the 1890s possessed antipyretic and analgesic activity like all other aspirin-like drugs but, unlike most, had little anti-inflammatory activity and was virtually devoid of gastric or platelet side effects. Apparently in accord with its therapeutic profile, paracetamol was found to have a different pattern of inhibitory activity, being more effective against crude brain COX preparations than those prepared from peripheral tissues such as the spleen. At the time (1972), this observation was put forward as a putative explanation for the selectivity of its therapeutic action and the idea that there were several forms of the enzyme was formulated. Wide variations in the inhibitory potency of indomethacin against COX enzymes prepared from a range of tissues was subsequently reported and the "isoenzyme" idea was further elaborated in several early reviews.

DISCOVERY OF OTHER COX ISOFORMS

Despite this early, presumptive, evidence for alternative forms of COX, little hard evidence emerged for

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several years probably because technical problems delayed progress. The structural features of COX, which has dual hydroperoxidase and COX activity, were not well understood and with the notable exception of the seminal vesicles of some species, the enzyme is usually expressed in tissues in low abundance. As a dimeric membrane-bound protein, COX posed many challenges to purification and was not sequenced until 1988. Other factors which caused confusion included the wide variations in assay conditions used by different laboratories as well as differences in the kinetics of the COX inhibitors. Many of these issues are still relevant today.

The 1980s, however, saw some exciting advances in our understanding of COX pharmacology. Morrison et al. [1] observed that the inflamed kidney unexpectedly developed an enormous capacity to generate prostaglandins that was due to de novo synthesis of fresh enzyme and in 1982, the group published evidence suggesting the presence of two distinct forms of COX in brain tissue with differing sensitivities to indomethacin. Other studies in GI tissue also supported the selectivity of action of NSAIDs in different tissues.

By the end of the 1980s it was clear that exposure of cells of various types to growth factors, cytokines, endotoxin, or some hormones induced the de novo synthesis of enzyme which, when partially sequenced, strongly resembled the bovine seminal vesicle COX.

While investigating the expression of early-response genes in fibroblasts transformed with Rous sarcoma virus, Simmons and his colleagues [2] identified a novel mRNA transcript that coded for a protein that was not identical but which had a high sequence similarity to the seminal vesicle COX enzyme and suggested that a COX isozyme had been discovered. Herschman and colleagues [3] also observed a novel cDNA species encoding a protein with a predicted structure similar to COX-1, while studying phorbol-ester–induced genes in Swiss 3T3 cells.

Masferrer et al. [4] identified an inflammation-inducible form of COX as the species that both Simmons and Herschman had cloned and the two COX enzymes were renamed *COX-1*, referring to the original enzyme isolated from seminal vesicles and subsequently found to be distributed almost ubiquitously, and *COX-2*, denoting the "inducible" form of the enzyme (although it was expressed basally in the brain and elsewhere). COX-1/COX-2 mRNA was differentially expressed in human tissues and promoter analysis confirmed a fundamental difference between the two isozymes, with the COX-2 promoter containing elements characteristic of genes that are switched on during cellular stress and downregulated by glucocorticoids, whereas COX-1 appeared to be a "housekeeping" gene.

WHICH ISOFORM IS MOST RELEVANT TO INFLAMMATION?

Histological and other studies confirmed this apparent division of labor between the two enzymes, showing, for example, that COX-1 seemed to be the predominant isoform in healthy GI tissue from rat, dog, and monkeys whereas COX-2 was seen mainly in inflamed tissues. This suggested that COX-2 was the predominant inflammatory species, and by implication, the best target for NSAIDs. So was COX-2 inhibition the true therapeutic modality of NSAIDs? If so, COX-1 inhibition might account for the side effects such as gastric irritation and depression of platelet aggregation. This was the notion put forward by more than one group and which came to be known as the "COX-2-bad:COX-1-good" hypothesis. If true, then the obvious corollary was that a selective COX-2 inhibitor should be an ideal drug, possessing anti-inflammatory actions but lacking gastric and other side effects.

At the time, the sequence of the catalytic domains of the two isozymes appeared to be so similar that the prospect of finding a specific inhibitor initially seemed remote and most of the nonsteroidal drugs available at that time were found to inhibit both enzymes to a greater or lesser degree. However, some selectivity of action was seen with experimental drugs such as 6-MNA, BF389, and DuP697.

An experimental compound (DuP697), produced by the Dupont Company in 1990, was observed to be an effective anti-inflammatory agent with reduced GI toxicity. Significantly, it showed only feeble activity in vitro using the COX-1 enzyme, but was more effective against rat brain COX preparations. The authors originally attributed the GI safety of this compound to its chemical structure (a nonacidic thiophene), which was presumed to produce a different pharmacokinetic profile to other COX inhibitors. However, other experimental compounds with similar properties were subsequently discovered and in the light of the discovery of COX-2, it was not long before it was realized that these drugs might have exhibit unusual behavior because they acted predominantly on the COX-2 isozyme. The two companies who were the first to exploit these ideas and to market the new "coxibs" were Searle Monsanto (now Pfizer) and Merck. The former focused on sulphonamide-substituted 1,S-diaryl pyrazole compounds that lead eventually to the discovery of celecoxib, whereas Merck scientists settled on a series of methylsulphonylphenyl compounds which, in turn, culminated in the manufacture and introduction of rofecoxib (since withdrawn).

In support of the COX-1/COX-2 concept, the pharmacology of some other selective COX-2 inhibitors were described in the literature. SC558 (a celecoxib

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prototype), for example, showed good efficacy in rodent models of inflammation, fever, and pain, whereas the structurally related SC560, a selective COX-1 inhibitor, was ineffective. Celecoxib itself (then known as SC58635) reversed carrageenan-induced hyperalgesia and local prostaglandin production in rats, and a related compound was active intrathecally.

The meloxicam (Boehringer Ingleheim) and etodolac (Wyeth/Shire) were already under development during the time that progress was being made with COX-2 biology, and nimesulide (Helsinn) had been on the market in Europe since 1985. In each case, evidence already indicated that these agents were different from other NSAIDs in terms of their GI tolerance and these drugs turned out to be effective COX-2 inhibitors. In an authoritative survey, Warner et al. (1999) found, for example, that meloxicam and etodolac showed almost the same order of selectivity for COX-1/COX-2 as some of the newer agents.

BIOCHEMISTRY OF COX ENZYMES AND MECHANISMS OF NSAID INHIBITION

Both COX isoforms are bifunctional enzymes, having two distinct catalytic activities: the first, *dioxygenase* step incorporates two molecules of oxygen into the arachidonic (or other fatty acid substrate) chain at C-11 and C-15 giving rise to the highly unstable endoperoxide intermediate PGG_2 with a hydroperoxy group at C-15. A second *peroxidase* function of the enzyme converts this to PGH_2 with a hydroxy group at C-15, which can then transformed in a cell-specific manner by separate *isomerase*, *reductase*, or *synthase* enzymes into other prostanoids. Both COX-1 and COX-2 are heme-containing enzymes which probably exist as homodimers attached to intracellular membranes.

Most NSAIDs inhibit only the initial dioxygenation reaction. They are generally "competitive-reversible" inhibitors but there are differences in their kinetics. Generally, these drugs inhibit COX-1 rapidly but the inhibition of COX-2 is time-dependent and often irreversible, as explained earlier. To block the enzymes, NSAIDs must enter the hydrophobic channel forming hydrogen bonds with an *Arg* residue at position 120, thus preventing substrate fatty acids from entering into the catalytic domain. The solution of the crystal structures of COX-1 in 1994 and COX-2 in 1996 made a substantial impact on the drug discovery effort. Despite their high homology, detailed examination of the structure of the catalytic sites revealed the substrate binding "channel" in the two enzymes to be quite different. A single amino-acid change, from the comparatively bulky *Ile* in COX-1 to *Val* at position 523 in COX-2 (equivalent to position 509 in COX-1), and the conformational changes that this produced resulted in enhanced access to a "side pocket" that allowed the binding of COX-2-selective inhibitors by providing a docking site for the phenylsulphonamide residue of drugs such as SC558. In an elegant demonstration of how crucial this single change was, Gierse et al. (1996) showed that mutation of this residue in COX-2 back to *Ile* largely prevented selective COX-2 inhibitors from working (see Figure 18.2).

These structural data also helped explain differences in the inhibitory kinetics of COX-1 and COX-2. There are, as stated earlier, several distinct mechanisms by which COX-1 inhibitors can inhibit the enzyme, but many are of the competitive-reversible type. By contrast, the COX-2 inhibitors are irreversible, time-dependent inhibitors, partly as a result of the binding of the sulphonamide (or related) moiety into the enzyme "side pocket." An analysis of the kinetic behavior of several COX inhibitors subsequently discerned four separate modes of enzyme inhibition, ranging from the simple competitive inhibition of drugs such as ibuprofen, through the "weak binding, time-dependent" mechanism of naproxen and the oxicams and the "tight binding, time-dependent" inhibition of indomethacin, to the covalent modification produced by aspirin. Alone among the NSAIDs, aspirin irreversibly acetylated a serine residue (*Ser* 530) within the COX active site thus blocking substrate access and inhibiting catalytic activity.

RELATIVE POTENCIES OF NSAIDS ON COX ENZYMES

Most "traditional" NSAIDs are inhibitors of both isoenzymes, though they vary in the degree to which they inhibit each isoform. It is believed that the anti-inflammatory action (and probably most analgesic actions) of the NSAIDs is generally related to inhibition of COX-2 whilst their unwanted effects – particularly those affecting the GI tract – are largely a result of their inhibition of COX-1. A broad scheme for classifying the relative selectivity for COX-1/2 of the currently available NSAIDs is given in Table 18.1.

ASPIRIN IN NONINFLAMMATORY CONDITIONS

Aspirin – previously thought of as an old anti-inflammatory workhorse – is now approaching the status of wonder drug that is not only of benefit in inflammation, but in an increasing number of other conditions. Most significant is its role in the therapy of *cardiovascular disorders*. The long-lasting antiplatelet action of low-dose aspirin may be use prophylactically to great clinical benefit in patients at high risk from (for example) myocardial infarction. Other diseases in which the drug is being (or has been) trialed include *colonic and rectal cancer*, *Alzheimer's disease*, and *radiation-induced diarrhea*.

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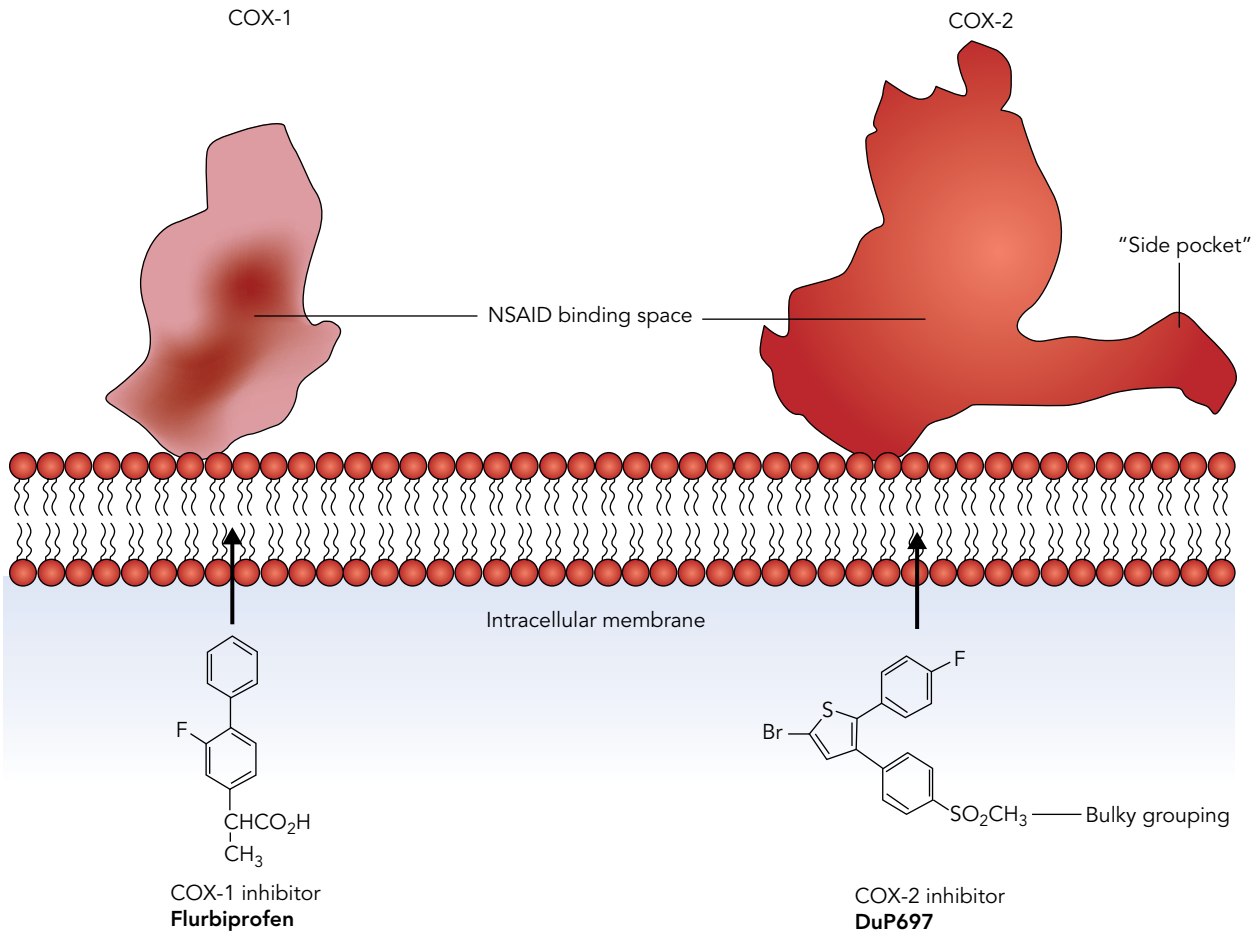


Figure 18.2. Comparison of drug binding sites in COX-1 and COX-2. This schematic diagram (modified from Flower 2003) shows how the presence of a "side pocket" allows access to the hydrophobic channel in the COX-2 enzyme, of inhibitory compounds (exemplified here by the drug DuP697), that cannot gain access to the channel in the COX-1 isoform.

PARACETAMOL

Whilst it is an excellent analgesic and antipyretic, the anti-inflammatory activity of paracetamol is very low and seems to be restricted to a few special cases (e.g., inflammation following dental extraction). As already mentioned paracetamol potently reduced prostaglandin synthesis in the central nervous systems (CNS), but not in the periphery. In experimental animals, analgesia and antipyresis with paracetamol were accompanied by the reduction in prostaglandin synthesis in the CNS. As paracetamol is a weak inhibitor of COX-1 and COX-2 activities, its pharmacological actions cannot be explained by inhibition of these enzymes. COX-3 identified as a splice variant of COX-1 in 2002 in canine tissues was shown to be inhibited by paracetamol. It was thus hoped that the discovery of COX-3 might provide a neat explanation for its pharmacological actions. However, the existence of COX-3 in rodent and human tissues is disputed because retention of intron-1 results in an out-of-reading frame sequence, despite some evidence in the literature on its possible expression in these species. A COX-1 variant, which may be related, is involved in thermoregulation in normothermic animals. Whilst being a weak anti-inflammatory drug, paracetamol produced the same effect as NSAIDs at the latter phase of acute inflammation, in that it prevented resolution of the inflammatory reaction (our unpublished observations).

SUMMARY

The NSAID field has undergone major transformation since the mechanism of action of these drugs was discovered, but what of the future? At the time of writing, the unique role of COX-2 selective inhibitors in therapy remains to be fully clarified and the propensity of the entire range of NSAIDs to cause hypertension requires elucidation. It is ironic that the precise mechanism of paracetamol action continues to elude us, as indeed does the mechanism of action of salicylate itself which is the major metabolite of aspirin and which has anti-inflammatory actions in its own right. Clearly there is

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TABLE 18.1. Relative inhibitory potency of some common NSAIDs and Coxibs on the two COX isoforms

Description	Selectivity ratio	Examples
Highly COX-1 selective	100–1,000	Ketorolac
Very COX-1 selective	10–100	Flurbiprofen
Weakly COX-1 selective	1–10	Indomethacin, aspirin, naproxen, ibuprofen
Nonselective; full inhibition of both isoforms	1	Fenoprofen
Nonselective; incomplete inhibition of both isoforms	1	Salicylate
Weakly COX-2 selective	1–10	Diflunisal, piroxicam, meclofenamate. Sulindac, diclofenac, celecoxib
Very COX-2 selective	10–100	Valdecoxib*, etoricoxib
Highly COX-2 selective	100–1,000	Rofecoxib*

Source: Warner and Mitchell, 2004.
*These drugs have been withdrawn from clinical use and are shown here for illustration purposes only.

much work still to be done and quite possibly further COX or COX-like enzymes to discover.

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19 Cytokines and Chemokines in Inflammation and Cancer

Thorsten Hagemann and Toby Lawrence

Cytokines and chemokines are peptide mediators that regulate a broad range of processes involved in the pathogenesis of inflammatory diseases and cancer. It is well established that an imbalance cytokine or chemokine activities can favor chronic inflammation leading to organ failure. Chemokines and cytokines are also implicated in malignant disease with links to tumor progression, angiogenesis, and invasion. Biological therapies targeting cytokines and chemokines have already improved outcomes of inflammatory disease and clinical trials are ongoing in cancer patients. Targeting tumor necrosis factor (TNF)-α represents a major success story for this approach. Anti-TNF-α was the first antibody against an inflammatory cytokine demonstrated to be efficacious in human disease and it showed over the years to be effective in a range of inflammatory diseases such as rheumatoid arthritis (RA), inflammatory bowel disease (IBD), and more recently, cancer.

CANCER AND INFLAMMATION

The cells and mediators of inflammation also form a major part of the tumor microenvironment. In some cancers, inflammatory conditions precede development of malignancy; in others, oncogenic changes drive a tumor-promoting inflammatory milieu. Whatever its origin, this "smoldering" inflammation aids proliferation and survival of malignant cells, angiogenesis, and metastasis; subverts adaptive immunity, and alters response to hormones and chemotherapeutic agents [1,2]. The cytokine (Figure 19.1) and chemokine network (Figure 19.2) is of great importance in the processes of cancer-related inflammation regulating both host and malignant cells in the tumor microenvironment [3]. The link between inflammation and cancer was first suggested by Rudolph Virchow in 1863, when he demonstrated leukocytes in neoplastic tissue. Virchow's original hypothesis has been revisited by many research groups [1]. The epidemiological data available are very impressive and show a clear association between chronic inflammatory conditions and subsequent malignant transformation in the inflamed tissue (Table 19.1).

CYTOKINES IN INFLAMMATION AND CANCER

It is known that neoplastic cells often overexpress proinflammatory mediators including proteases, ecosanoids, cytokines, and chemokines. Several cytokines, such as TNF-α, IL-6, IL-17, IL-12, IL-23, IL-10, and TGF-β, have been linked with both experimental and human cancers and can either promote or inhibit tumor development. Here we will describe some of the cytokines that may be potential targets for cancer therapy.

TNF-α is the prototypical proinflammatory cytokine, the critical role of TNF-α in chronic inflammatory diseases is well known [4], although originally shown to be toxic to tumor cells in high doses (hence tumor *necrosis* factor) the tumor-promoting function of TNF-α have been clearly demonstrated in mice [5]. TNF-α can be produced by malignant cells or inflammatory cells in the tumor microenvironment and TNF-α signaling can promote cell survival, invasion, and angiogenesis [6]. In mesothelioma phagocytosis of asbestos fibers by macrophages leads to TNF-α release that promotes cell survival and thereby reduces asbestos-induced cytotoxicity [7]. TNF-α has also been proposed to contribute to tumor initiation by stimulating the production of genotoxic reactive nitrogen (RNS) and oxygen (ROS) species [8]. Genetic polymorphisms that enhance TNF-α production are associated with increased risk of multiple myeloma (MM), bladder cancer, hepatocellular carcinoma (HCC), gastric cancer, and breast cancer, as well as poor prognosis in various hematological malignancies [9]. Other actions of TNF-α that might promote tumorigenesis, as opposed

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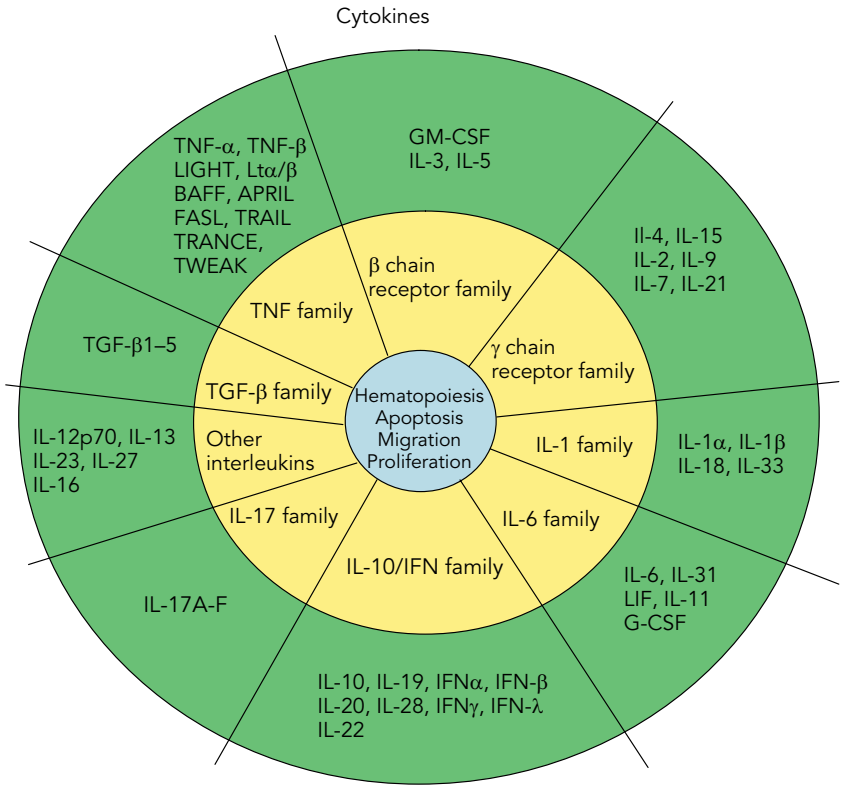


Figure 19.1. The cytokine network.

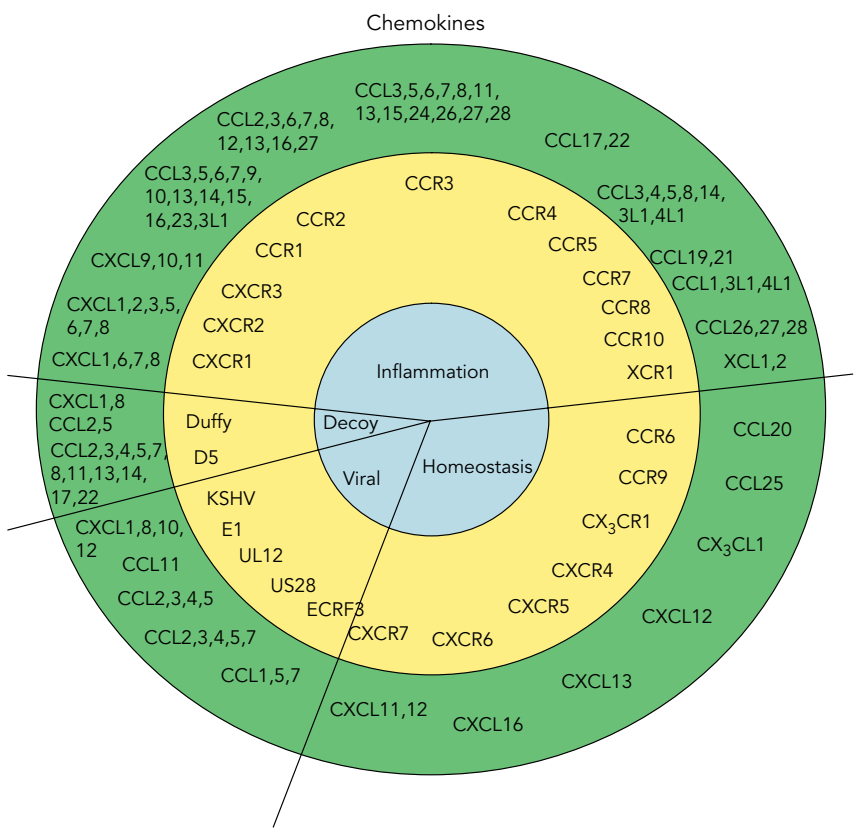


Figure 19.2 The chemokine network.

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TABLE 19.1. The association between inflammation and cancer in different organ systems

System	Inflammation
Respiratory	Asbestos and mesothelioma silica, Cigarette smoke and bronchial cancer Chronic asthma and bronchial cancer
Genitourinary	Human papilloma virus and penile carcinoma Schistosomiasis and bladder cancer
Reproductive	Pelvic inflammatory disease, ovarian epithelial inflammation, and ovarian cancer
Gastrointestinal	Epstein–Barr virus and nasopharyngeal cancer Barrett's metaplasia and esophageal cancer *H. pylori* gastritis and gastric cancer Chronic pancreatitis and pancreatic cancer Chronic cholecystitis and gallbladder cancer Hepatitis and hepatocellular cancer Inflammatory bowel disease, and colorectal cancer Human papilloma virus and ano-genital cancer
Hematopoietic	Epstein–Barr virus and Burkitt's lymphoma HTLV1 and T-cell leukemia/lymphoma

to tumor initiation, include promotion of angiogenesis and impairment of immune surveillance through T-cell suppression and inhibiting the cytotoxic activity of activated macrophages [10].

TNF-α binds to two receptors, the ubiquitously expressed TNFR1, and TNFR2, which is restricted to expression on hematopoietic cells. Previous studies show that TNFR1 is important in tumor promotion. TNFR1 knockout mice are resistant to skin carcinogenesis [11], experimental lung metastases in the RENCA renal cancer model [12], and experimental liver metastases were attenuated in TNFR1-deficient mice [13]. In chimeric mice whose bone marrow was repopulated with TNFR1–/– cells, development of colitis and colon cancer was attenuated [14]. Several studies have demonstrated that stromal cell TNF-α is tumor promoting. In a genetic model, bone marrow cell TNF-α was implicated in promotion of inflammation-associated liver tumors [15]. In a model where chemical damage led to liver cancer, Kupffer cell TNF-α was one of the mitogens driving proliferation of genetically damaged hepatocytes [16]. In a chemically induced model of colitis and colorectal cancer, mononuclear cell TNF-α was implicated in inflammation and subsequent tumor development [14]. Several studies have also reported therapeutic activity of anti-TNF-α antibodies, or a TNF receptor fusion molecule, in genetic models of liver and colorectal cancer [15,17]; a carcinogen-induced model of colorectal cancer [14]; and pancreatic cancer xenografts [18], although the exact mechanisms of action are not understood.

In many different experimental and human cancers, malignant cells produce TNF-α during tumor growth and spread [5,6,18–24]. Preclinical experiments with TNF-α antagonists and early phase clinical trials of TNF-α antagonists in patients with advanced cancer suggest that this inflammatory cytokine may be a useful target [25–28].

IL-6 is a pleiotropic inflammatory cytokine that is considered a key growth factor for both malignant and immune cells. Most IL-6 target genes are involved in cell cycle progression and suppression of apoptosis, which underscores the importance of IL-6 in cell survival and tumorigenesis. IL-6 is suggested to have a pivotal role in the pathogenesis of Castleman's disease [29] and multiple myeloma (MM) [30]. Clear evidence that IL-6 governs the growth of MM, a malignant disorder of plasma cells, has come from studies using IL-6 knockout mice, which were found to be resistant to plasmacytoma induction [30]. In MM, IL-6 is produced by stromal cells in the bone marrow and its synthesis by these cells can be further enhanced by their interaction with malignant plasma cells. New IL-6 antagonists are being evaluated for treatment of MM such as small molecules and monoclonal antibodies. IL-6 is also a key mediator of inflammatory disease, IBD and colitis is associated with high concentrations of IL-6 [31], in experimental models of colitis-associated colon cancer (CAC) and HCC IL-6 production by myeloid cells is also critical for carcinogenesis [32–34].

IL-10 is an immunosuppressive and anti-inflammatory cytokine also linked with inflammation-associated cancer [35]. IL-10-deficient mice develop spontaneous colitis, due to hyperactivation of immune cells, and eventually CAC. Expression of IL-10 by tumor cells and macrophages is also thought to promote the development of Burkitt's lymphoma through the production of the TNF family member BAFF, which promotes B cell and lymphoma survival [36]. An elevated amount of IL-10 in the plasma has been correlated with poor prognosis in diffuse large B cell lymphoma patients [37]. In addition to direct growth modulation of malignant cells, the ability of IL-10 to suppress adaptive immune responses has also been suggested to favor tumor escape from immune surveillance [38]. Therefore, IL-10 has complex effects on tumor development. In some experimental systems, IL-10 is found to exert antitumor activity, but in other cases it can be promote tumorigenesis.

Recently, a new T cell subset named "Th17" characterized by the production of IL-17 was identified as an important player in inflammatory responses [39]. IL-17 induces the recruitment of immune cells and leads to the induction of many proinflammatory factors, including TNF-α, IL-6, and IL-1β [40]. Studies have begun to

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address the role of IL-17 in chronic inflammation and cancer [41]. IL-17 overexpressing human cervical cancer cells and nonsmall cell lung carcinoma cells show substantially greater ability to form tumors in immunocompromised mice compared with control cells not overexpressing IL-17 [42,43]. In primary NSCLC samples, IL-17 expression has frequently been detected in tumor-infiltrating inflammatory cells and was associated with increased tumor vascularity [44]. However, there is also evidence that IL-17 might be involved in tumor surveillance in immunocompetent mice [45].

This presents a caveat for cytokines as targets in cancer therapy, although inflammation is clearly linked with tumorigenesis, both the innate and adaptive immune systems have the capacity to recognize and eliminate malignant cells [46,47]. Many proinflammatory cytokines may function in tumor immune surveillance and it is vital to determine whether the function of potential therapeutic targets, such as IL-17, contributes to antitumor immunity.

CHEMOKINES IN INFLAMMATION AND CANCER

Chemokines are chemotactic cytokines that regulate cell trafficking in development and disease. In inflammation chemokines regulate leukocyte recruitment, diapedesis, activation, and retention within the inflamed tissue. Other chemokines, such as CCL2, CCL5, CCL21, and CXCL13, control lymphoid organogenesis. Chemokines also regulate the retention and release of stem cells from the bone marrow. In all these activities, chemokines function together with cytokines that regulate their expression and that of their reciprocal receptors. Chemokine expression and their functional importance in inflammation have provoked several therapeutic targeting studies, although with limited success so far [46].

Chemokine receptors and their ligands influence cell motility, invasion, and survival in experimental models of cancer [3]. Malignant cells and tumors express a large number of chemokines and chemokine receptors; accordingly, tumor cells are able to recruit leukocytes and endothelial or mesenchymal cell progenitors from the circulation into the tumor microenvironment. Many chemokines are produced downstream of oncogenic mutations; Ras mutations induce CXCL1 (Gro-1) [48], c-myc activation induces CC chemokines [49], and the RET/PTC gene fusion induces several chemokines including CXCL12 [50].

These chemokines control the extent and phenotype of the leukocyte infiltrate in cancer [51–53]. Ablation of leukocyte function, their infiltration into tumors [54], or switch of their phenotype [55] can inhibit tumor growth and metastases.

Until recently, the evidence the role of chemokines in carcinogenesis was largely inferred from correlative studies and experiments with antagonistic antibodies. Genetic evidence for a tumor-promoting role of chemokines eventually came from the discovery of the D6 decoy receptor that functions to block chemokine function [56]. D6 null mice are inefficient in removing chemokines from inflamed sites and therefore show exaggerated responses to otherwise innocuous inflammatory agents [56]. These mice were also recently shown to have increased susceptibility to skin tumors [57].

There are at least 40 different human chemokines and 20 chemokine receptors [58]. They are divided into CC, CXC, XC, and CX3C subfamilies according to the nature of the conserved cysteine motif [59]. Chemokines bind to seven transmembrane-spanning (7TM) receptors typically coupled to G proteins for signaling [60]. There are 11 CC chemokine receptors, seven CXC chemokine receptors, and single receptors for XC and CX3C chemokines [59,61]. These can be broadly subdivided into constitutive and inflammatory [59,61]. There are two nonsignaling receptors, Duffy and D6, thought to attenuate the actions of inflammatory chemokines [62], and viral chemokine and chemokine receptor homologues [63] (Table 19.2).

Positive preclinical data in animal cancer models show that chemokine/receptor antagonists reduce the leukocyte infiltrate into tumors, leading to reduced angiogenesis and cancer growth, and that targeting chemokine receptors on malignant cells inhibits survival, metastasis, and peritoneal dissemination (Table 19.3).

Therapeutic antibodies are likely to provide the most successful way to inhibit the chemokine/chemokine receptor axis. Anticytokine antibodies have had clinical success in inflammatory diseases [70–73]

TABLE 19.2. Drugs targeting chemokines or their receptors currently in clinical development or recently licensed

Target	Disease	Stage
CCR5	CCR5-trophic HIV	Licensed
CXCR4	Stem cell mobilization in cancer	Licensed
CXCR4	Metastatic cancer	Phase I/II
CCR9	Crohn's disease	Phase II
CCR4*	Adult T cell leukemia, severe asthma	Phase I
CXCR2	COPD	Phase II
CCL2*	Advanced cancer and COPD	Phase I

*Therapeutic antibody.
Abbreviation: COPD, chronic obstructive pulmonary disease.

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TABLE 19.3. Targeting chemokines and their receptors in mouse cancer models

Target	Cancer model	Action	Reference
CCR5	Breast cancer	Inhibited leukocyte infiltrate and tumor growth	64
CCL2*	Prostate cancer	Inhibited tumor growth and metastasis	65
CC chemokines	Skin cancer	Inhibited leukocyte infiltrate and tumor development	57
CXCR4	Lymphoma	Inhibited tumor growth	66
CXCR4	Glioma	Inhibited tumor growth	67
CXCL8	Ovarian cancer	Inhibition of tumor growth	68
CXCR7	Breast and lung cancer	Inhibition of angiogenesis and tumor growth	69

*Therapeutic antibody.

TABLE 19.4. Antibody therapeutics in the clinic

Target	Disease	Stage
TNF-α	Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease	Marketed
B Lys	Systemic lupus erythematosus, rheumatoid arthritis	Phase II, III trial
IL-12	Crohn's disease, psoriasis	Phase II trial
TRAIL-R1	Variety of tumors	Phase II trial
TRAIL-R2	Advanced solid tumors	Phase II trial
IL-13	Severe asthma	Phase I trial
TGF-β	Idiopathic pulmonary fibrosis	Phase I trial
GDF-8	Muscle wasting disease	Phase I trial
Eotaxin*	Allergic conjunctivitis	Phase I trial

* Eoxtaxin is a chemokine (CCL11).

and there are promising preclinical [15,18,74,75] and early-phase clinical trial results in cancer [26–28]. Antibodies against chemokine receptors are likely to have significant advantage because of the redundancy of chemokine ligand binding to some receptors.

RISKS OF TARGETED THERAPEUTICS

The functions of cytokines and chemokines in normal physiology have an important role in the healthy host. Foremost, TNF, IL-1, and IL-6 have critical roles in host defense in both the innate and adaptive immune systems. Therefore, one should expect an increase in the rates of infections, which is sometimes the case [76,77]. For example, TNF plays a pivotal role in granuloma formation and, therefore, in the defense against intracellular pathogens. Indeed, reactivation of tuberculosis has been observed with TNF inhibitors. In contrast, IL-6 may have an inhibiting role on granuloma formation [78]; therefore, targeting IL-6 therapeutically might not induce reactivation of, or promote infections with, some intracellular pathogens. The side effects of these targeted therapeutics on host defense may be particularly important in cancer patients who may already have seriously compromised immune systems.

Other side effects have been reported with IL-6 inhibition, such as increases in plasma lipid levels, but it remains to be seen whether cardiovascular risk increases. IL-6 is also a growth factor for hepatocytes, and tocilizumab (IL-6R neutralizing antibody) leads to increases in hepatic enzymes, which are usually of transient nature and not associated with hepatitis [77]. However, long-term follow up will have to show whether this might be associated with liver damage. Given that TNF inhibition also leads to normalization of IL-6 [79] but not liver enzyme elevations, it will also be of interest to learn whether a monoclonal anti-IL-6, rather than an antibody to the IL-6R, will induce a similar adverse event profile. Although antibodies offer exquisite specificity and high avidity, often these biological agents are chimeric or humanized monoclonal antibodies, therefore sensitization might occur, which can lead to both allergic reactions and reduction in efficacy. However, a number of candidate therapeutic antibodies (Table 19.4) have been identified and successfully tested in the clinic.

FUTURE DIRECTIONS

There are many additional molecules, both cytokines and chemokines, that may be future targets for effective therapy in inflammation and cancer we have merely highlighted a few. Targeting of pro-inflammatory cytokines such as TNF-α or IL-6 is highly efficacious in inflammatory disease such as RA and,

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at least for anti-TNF agents as well as other chronic inflammatory disorders. There are also very positive early signs for anti-TNF therapy in cancer. It is clear the full potential of these agents in cancer therapy may only be realized in combination with tolerable regimes of existing chemotherapy. The challenge now is to identify a rationale for which combinations will be most effective in which patients. Further mechanistic studies on the role of cytokines and chemokines in carcinogenesis will undoubtedly reveal new potential targets with increased efficacy and reduced side effects.

Cytokines are structurally diverse molecules with diverse and cell type–specific activities. Cytokine functions are mediated by binding specific receptors and their activities include regulating cell activation, hematopoiesis, apoptosis, migration, and proliferation. By virtue of their broad spectrum of activity in the hematopoietic system, they are implicated in virtually all aspects of innate and adaptive immunity. White blood cells are the primary source of cytokines, although they may also be produced by other cell types in the context of disease. This figure illustrates the scope of the cytokine network grouped into families of related molecules (Figure 19.1).

Chemokines are a large superfamily chemotactic cytokines that function to regulate leukocyte trafficking and activation. Chemokines are divided into subfamilies based on conserved amino acid (CC, CXC, CX_3C) sequence motifs. Most family members have at least four conserved cysteine residues that form two intramolecular disulfide bonds. The subfamilies are defined by the position of the first two cysteine residues. The inflammatory chemokines regulate leukocyte recruitment during inflammation, allergic responses, infection, and autoimmune diseases. There are also constitutive chemokines that regulate cell migration in homeostasis, hematopoiesis, and angiogenesis. Both inflammatory and homeostatic chemokines have been implicated in tumor growth and metastasis (adapted from F Balkwill, *Nat Rev Cancer* 2004) (Figure 19.2).

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